



HORIZON

ISP infrastructure implementation

By

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Project Thesis

Date:14/12/2024





HORIZON

Copyright





Declaration

I hereby declare that this submission is my own work and that, to the best of my knowledge and belief, it contains no material previously published or written by another person nor material which to a substantial extent has been accepted for the award of any other degree or diploma of the university or other institute of higher learning, except where due acknowledgment has been made in the text.

Signature Ali Khalid **Name** Ali Khalid **Date** 10/13/2024

Abstract

In order to satisfy the increasing needs for security, performance, and dependability, this isp project aims to optimise Horizon Technologies' ISP network infrastructure. A variety of cutting-edge networking protocols both for horizon and for the clients and technologies were used in the project, with a focus on segmented network administration, secure communication, dynamic addressing, and redundancy. By configuring EtherChannels and adding backup routers for layer 3 routing , redundancy was created, guaranteeing connection dependability and reducing the possibility of network outages at horizon. To facilitate user and network authentication, a Windows server with a strong triple-layer security configuration was incorporated. Additionally, a DHCP server was used to dynamically assign IP addresses to speicifr branches that requested it, within Horizon's client network, improving manageability and efficiency in IP distribution, and a dedicated VPN was put in place to facilitate secure remote access and mail plus MPLS for QOS.

IPsec was used for encrypted communication over public networks, MPLS was used to support Quality of Service (QoS) and prioritise important traffic for special customers, and Border Gateway Protocol (BGP) was used for scalable and dependable inter-domain routing between isp companies. In order to improve network security and segmentatiosn, Virtual Routing and Forwarding (VRF) was also implemented to reduce costs for horizon. This



allows for the creation of separate routing tables for various client segments. Network Address Translation (NAT), which enables internal IPs to be mapped to public IPs, was set up to facilitate safe internet access for client devices. This ensures security and effective IP utilisation. To further strengthen network security and prevent unwanted access, Access Control Lists (ACLs) and improved port security mechanisms were added to these protocols.

Each component's operating was thoroughly verified during testing phases done by horizons technicians, which included NAT setup checks, IPsec connection stability, DHCP IP address distribution, MPLS and VRF performance evaluations, BGP route verification, and VPN access dependability. The outcomes showed notable gains in network scalability, security, and resilience for horizon. The project's results demonstrated the effective implementation of a flexible and safe ISP network design that matched Horizon's customers' changing needs in bahrian.

Future enhancement suggestions include adding more border routers to increase redundancy, incorporating DHCP server access authentication methods and deploying as a part of the normal package for clients, requiring encrypted passwords on all network devices, implementing OSPF authentication for internal routing security, and developing a management interface specifically for MPLS routers to simplify administrative duties. With a focus on security, performance, and adaptability, this project represents a major improvement in ISP network optimisation. VRF is emphasised as a crucial element in attaining network segmentation and overall security and HSRP can be added for more security.

Acknowledgements

First of all, I would like to express my sincere gratitude to Mr. Shahnawaz, my PMO for the project, for his important advice and valuable assistance during this project. His knowledge and precise outlook into every phase of the process, not only for the project but general academia, have been invaluable in assisting me in overcoming the obstacles and expanding my knowledge. My educational experience for the in-house project has been much enhanced by his organised methods and encouragement to independently investigate solutions. I sincerely appreciate the time and effort he invested in my growth, making sure I understood every stage completely.

I am also deeply appreciative of my supervisor, Ms. Sharifa, whose feedback and guidance have been fundamental to the success of this project. her being a networking specialist makes Her thoughtful advice and consistent support during our weekly meetings provided me with the tools to approach complex tasks with confidence. Her willingness to share her knowledge and her commitment to my growth for the in-house project have had a lasting impact on both my skills and my professional outlook.



Thank you both for your mentorship, encouragement, and dedication. I feel fortunate to have had the opportunity to work under your guidance and am inspired by the commitment you have shown to my progress.

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Introduction

Project rationale.

In a setting like the Kingdom of Bahrain where dependable, fast internet is essential, the goal of this project is to optimise the infrastructure of Horizon Technologies, a developing Internet service provider (ISP). The need for more redundancy, enhanced security, and effective resource management are among the issues Horizon faces that are typical of ISP networks. Slow connectivity, bad network uptime, and data security flaws are examples of performance problems brought on by



inadequate infrastructure or outdated technologies that can have a big influence on users and interfere with corporate operations.

In today's connected world, a robust ISP infrastructure is essential not only for user satisfaction but also for sustaining productivity, economic growth, and societal advancement for companies or clients. Poor network reliability and security can affect critical services such as government services or health and medical servers, prevent timely communications, and expose users to risks associated with cyber threats and data breaches.

This project seeks to directly solve these problems by putting sophisticated networking protocols, redundancy measures, and security improvements into place. The objective is to offer Horizon's customers a smooth, superior online experience that promotes their connectedness and helps the larger digital economy. This initiative will improve digital connection and help set a new benchmark for ISP service quality, which will benefit both enterprises and individual consumers.

Project objectives

The urgent need to strengthen Horizon's network against these threats and make it stand next to other local ISP's is what motivates this project, as it will increase its capacity to serve an expanding customer base while preserving dependable, fast, and secure internet services.

This project seeks to directly solve these problems by putting sophisticated networking protocols, redundancy measures, and security improvements into place both by training horizon employees and adding new network infrastructure and protocols. The objective is to offer Horizon's customers a smooth, superior online experience that promotes their connectedness and helps the larger digital economy. This initiative will improve digital connection and help set a new benchmark for ISP service quality, which will benefit both enterprises and individual consumers.

Prior Work

Previous research and applications pertaining to network infrastructure, multi-ISP connection, and ISP-specific security protocols done by horizon research department serve as the basis for this project. Many approaches of connecting across ISP networks in Bahrain and the Middle East have been thoroughly studied in the past, with an emphasis on protocol selection, redundancy alternatives, and routing strategies that preserve efficiency and security. Also, research on cybersecurity in ISP infrastructures offers information on methods for safeguarding data flow and keeping packets safe, between public and private networks or between different clients, such as VPNs, ACLs, and NAT.

While these sources contribute valuable knowledge, they often lack real-world, hands-on perspectives on integrating diverse network technologies within a single ISP's operational environment. Most prior work emphasizes either large-scale ISP infrastructures or focuses on



single-ISP systems without addressing the unique challenges of managing multiple connections across distinct ISPs like Horizon's network and ensuring seamless, secure internal communication. Furthermore, existing research may not cover the intricate implementation details involved in scaling security features, such as VRF for segmented routing or multi-layered server protection, within a flexible and adaptable network structure.

By converting the study team's theoretical discoveries into workable, configurations within Horizon Technologies' network, this method aims to close these gaps between theory and practical work. This project intends to offer useful advice for comparable implementations by documenting the integration procedure, protocol modifications, and security layers used in an actual ISP setting. This will give a nuanced understanding of how ISP networks can attain the best possible connectivity, resilience, and security in dynamic, multi-ISP environments.

Proposed Solution

With the goal of enhancing connectivity, security, and overall network performance for horizon technolgiess, this project offers a comprehensive solutions to the problems, with Horizon Technologies' ISP infrastructure. To build a robust, interconnected network, the approach entails combining sophisticated routing and switching configurations across Horizon's ISP connections and for clients both customers and companies. In order to provide effective and safe data sharing between ISPs, a major focus will be on using EtherChannel to offer redundancy and guarantee reliable connections, in conjunction with protocols like BGP for external connectivity and VRF for internal segmentation.

For secure communication, Virtual Private Networks will be established to protect sensitive data transmissions between ISPs and client branches who asked for extra servicess, ensuring confidentiality and data integrity for tge customers. In order to reduce the possibility of bottlenecks and improve network performance overall, the solution will internally optimise data flow utilising meticulously crafted subnetting and routing protocols that provide both scalability and redundancy.

The project intends to implement strong security measures externally to guard against potential threats and unauthorised access. These precautions, which together secure external interfaces and sensitive data flows, include firewall setups, Access Control Lists , and encryption methods. To strengthen the network's defence against intrusions and hackers.

Tasks

- ensure internal routing is working
- ensure external routing between isp is working
- ensure QOS is working for special clients
- insure network uptime for clients
- apply security to handle all attacks or threats on the network

Description of the report

This thesis offers a thorough review of the project that was carried out to improve the ISP infrastructure of Horizon Technologies in bahrain . The Background Work section lays the groundwork for comprehending the goals and extent of the suggested solution by introducing important technological ideas and providing an outline of the technologies in use for the project.

Critical elements including functionality, security, and scalability are addressed in the Requirements Engineering section, which describes the approaches used to determine, evaluate, and specify the project's needs in the isp. The strategies and decision-making procedures that were done by the project managment team that shaped the ISP network's development are explained in detail in the "Design and Methodology" section, with an emphasis on effectiveness, flexibility, and achieving project goals.

The Implementation section describes the importnat actions done to make the suggested solution a reality, such as setting up protocols, integrating devices, and putting strong security measures in place. The Testin" section, which details the processes and approaches used to evaluate performance, dependability, and security against predetermined targets as well as methods of tsting the hardware load, is essential to assessing the project's results.

The Discussion and Conclusion section concludes by summarising the findings in this thesis, emphasising the goals met, difficulties faced, and lessons learned doing this projectt. also making suggestions for future improvements at horizon technologies.

Background

Related theory

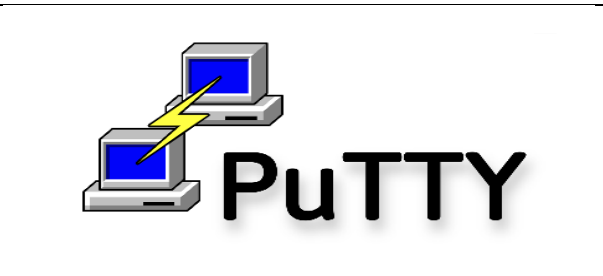
talking about related theory specially ISP in modern times, where persistent internet connection supports both personal and economic activity, an ISP network's performance and dependability are essential. ISPs encounter difficulties with insufficient security measures, ineffective IP address distribution, and a lack of redundancy.

By combining several physical lines, redundancy—achieved by technologies like EtherChannel—guarantees continuous communication and reduces downtime during failures. DHCP servers facilitate dynamic IP address management, which improves client connectivity and optimises resource allocation. Strong security measures are also necessary to defend against cyberattacks and threats both internal and external, such as ACLs for traffic control, VPNs for encrypted communication, and NAT for concealing internal IPs.

By combining advanced redundancy, IP management, and security protocols, this project helps Horizon Technologies overcome these obstacles and provide an ISP infrastructure that is scalable,

dependable, and secure and able to compete with other isp companioes not only in the middle east but globally .

Project technologies

	EVE-NG (Emulated Virtual Environment - Next Generation) is a powerful network emulation platform that enables users to design and simulate complex network topologies in a virtualized environment. It supports the emulation of a wide range of networking devices, such as routers, switches, and firewall s and pc's . For my horizon's, focused on implementing an ISP network infrastructure, EVE-NG was utilized due to its user-friendly interface and capabilities. After conducting thorough research, it seemed the best emulation software because of the experience gained in the last 3 years of networking
	VMware is a virtualization and cloud computing tool that uses virtualization software. They produce virtualization, networking, and security management tools and in the horizon project it was mainly used to run virtualization for eve-ng on the windows computer.
	PuTTY is an open-source terminal emulator, serial console and network file transfer application. It supports several network protocols, including SCP, SSH, Telnet, rlogin, it was used in the horizon project to enable the engineers to open up the router and switch command lines.

	WinSCP is a file manager, SSH File Transfer Protocol, File Transfer Protocol, WebDAV, Amazon S3, and secure copy protocol client for Microsoft Windows it was used in the horizon project to transfer computer images from the home computer to the Linux based eve-ng environment, it was used to transfer windows server 2019
	Chrome is web browser developed by google, other than it being used for online research for the horizon project, it was used to access the eve-ng GUI to gain access to the visual topology and wiring of the network

Related work and literature review

This section examines current approaches to improving ISP infrastructure globally and in the Middle East, with an emphasis on security and redundancy measures for the future.

For dependable service delivery, the network must have redundancy. Numerous studies have previously examined the use of multipath protocols, such EtherChannel, for link aggregation and even the addition of additional backup routers. These techniques and protocols enhance network fault tolerance and bandwidth utilisation. Other methods, like as load balancing and hybrid redundancy, have been proposed to lessen service disruption during equipment failures or traffic surges. However, a large number of these studies mostly explore the technology approaches without going into how they connect to more fundamental ISP infrastructure requirements, like horizon technologies, with client administration and service scalability.

So, Security and network segmentation are equally crucial for ISP networks. Research has shown that ACLs and VPNs are effective in protecting data and preventing unauthorised access in the majority of international IP companies. Additionally, if the ISP provides those services, methods like Virtual Routing and Forwarding (VRF) are being used to separate client networks and enhance security in shared ISP configurations. Although these techniques are helpful, they often lack integration with protocols like DHCP and NAT, which are essential for providing scalable, secure, and efficient client services.



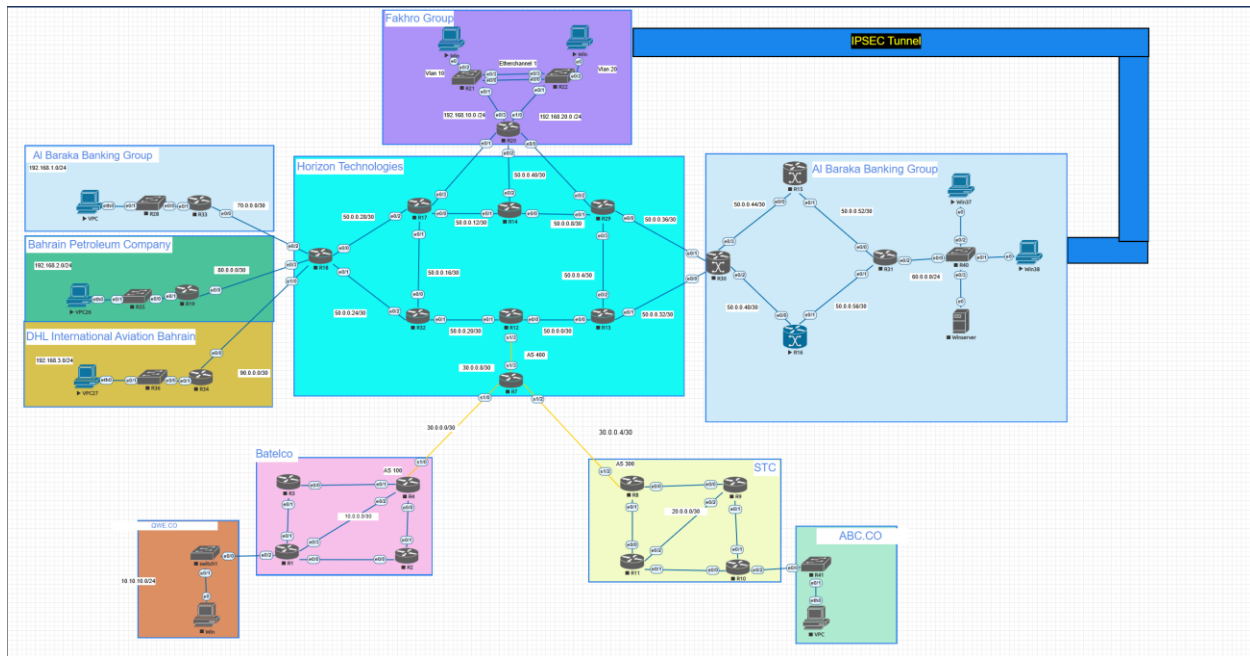
These studies are researched by the horizon research team are used in the horizon project to create a solution that is specific to Horizon Technologies and bahraini companies. It solves real-world issues while guaranteeing a scalable and resilient ISP infrastructure by using redundancy methods, strong security standards.

Methodology

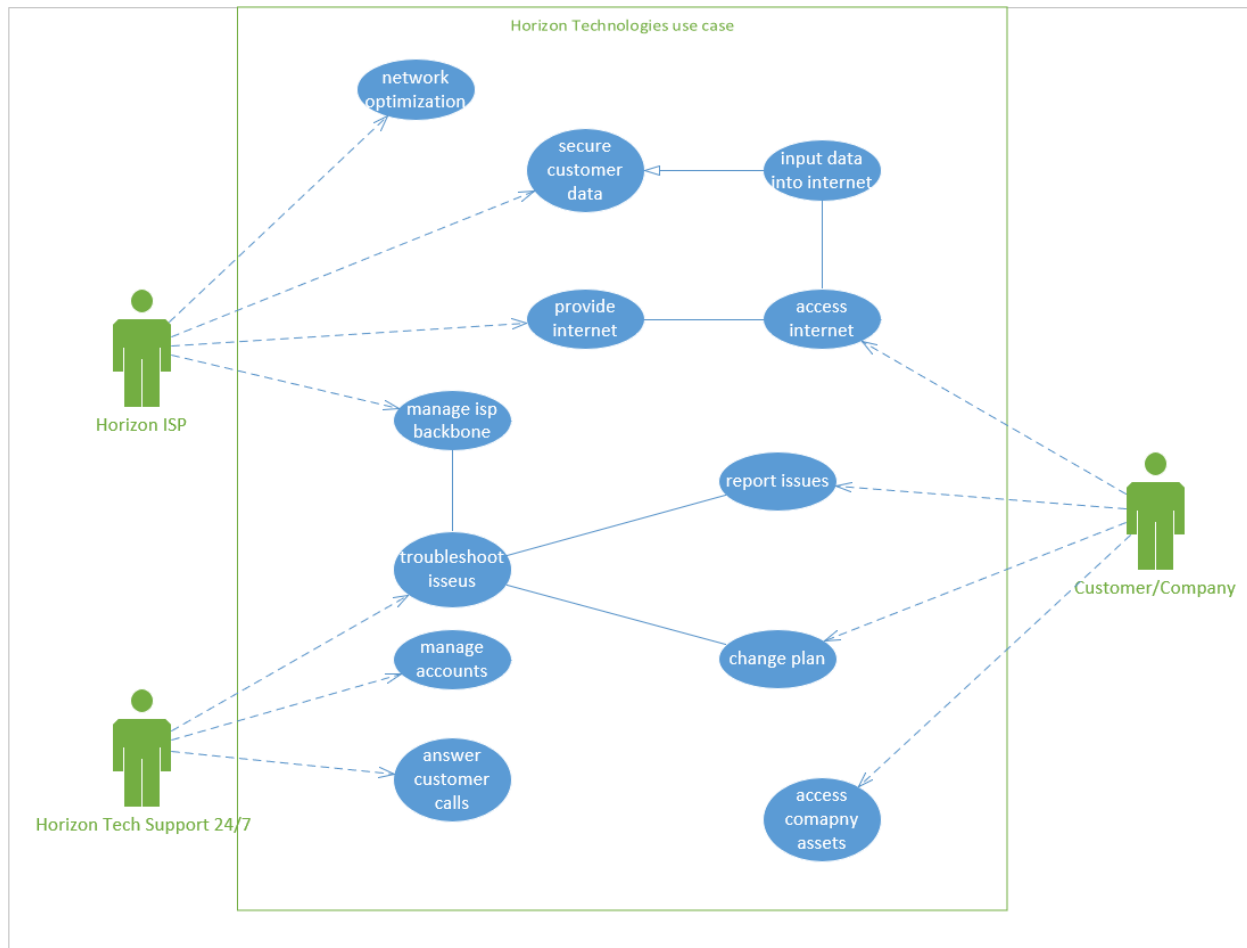
Solution design

So, when designing the new topology in the eve-ng for the horizon project the team had a few points that needed to be met, in a technological prospective, first of all, all the different isp needed unique AS numbers for the border gateway protocols, they all needed to have internal and external routing, each isp company needed to have their own client's provided with the network protocols requested, including security. as for Horizon, it was necessary to know which clients should be connected to which routers, based on location, which protocols were necessary based on requirements for the clients, some clients requested for example a vpn connection, some requested Qos, so a visual representation of the topology was created to get a clear understanding of the project requirements and different charts were designed to help give valuable insights into the horizon project

Solution design

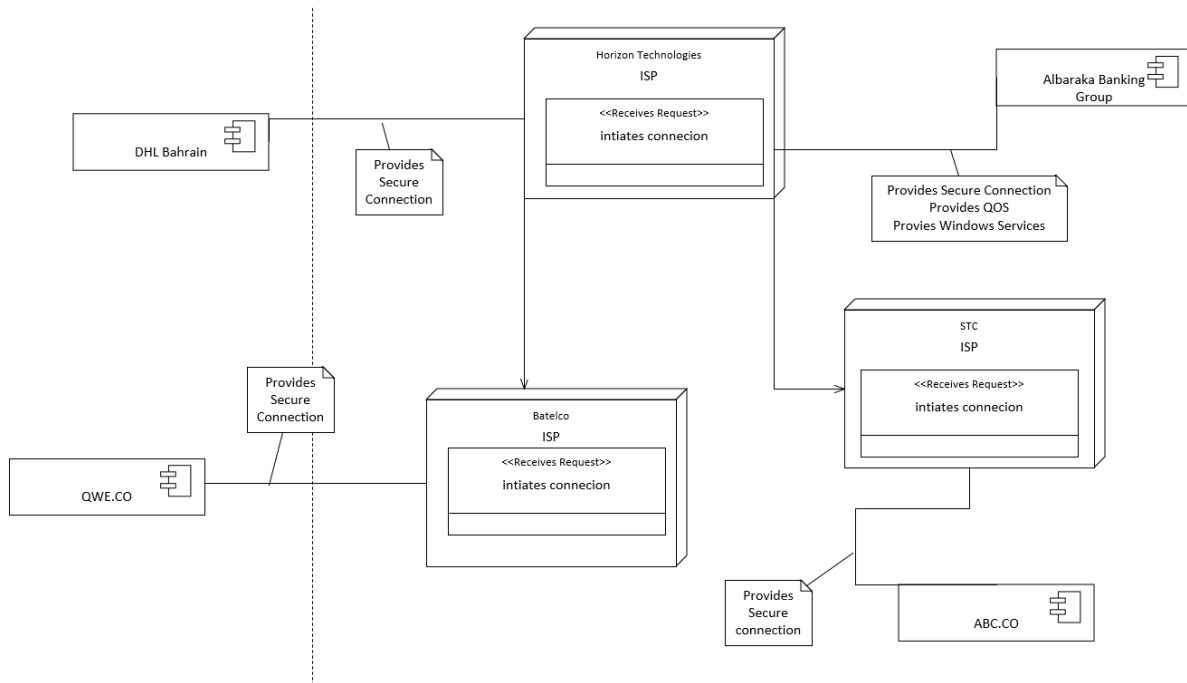


Use Case Diagram



The Use case Diagram above different actors who all have different jobs, Horizon ISP and tech support work together to comply with the customers/companies actors, they are mainly connected through the manage isp backbone link.

Deployment Diagram



The following deployment diagram shows the different connections between horizon technologies, Batelco and STC, and their different clients with requirements.

Interview plan

PROJECT No. and Title: Brief 06/ Horizon Technologies	PROJECT MANAGER: Ali Khalid
DATE CREATED: 9/27/2024	DATE LAST UPDATED: 10/12/2024
Interview Date and Time: 6 Pm 9/27/2024	Interviewee: Ali Khalid (Project Manager)

Question	Answer	Requirement
----------	--------	-------------

Do you need to collect additional information? If yes, can you give me an example?	Yes, we need to keep information about the current IP address allocations, VLAN configurations	Redesign the used form to include DHCP details, VLAN IDs, and performance indicators.
How often do you monitor network performance?	We monitor network performance daily	Set up regular performance reviews and reporting.
What security measures are currently in place?	We have firewalls and intrusion detection systems.	Review current security policies and ensure they are documented.
How is OSPF configured in the network?	OSPF is set up with area 0 as the backbone area, and additional areas are configured for branch offices.	Create a detailed topology diagram of the OSPF configuration and verify area settings.
Do you need bop from isp to isp	Yes, bgp is required	Implement BGP from border isp routers
Do you require QOS	Yes, it is required	Implement MPLS for special customers
Do you require vlans at layer 2 devices	Yes so, we can divide the departments	Implement vlans
Do you need a mail server, if you which type do you want	Yes, a mail server is required for clients who ask for them, any server is fine	Implement mail server
Do you require backup links from isp to isp	Yes.	Implement backup links
Is vrf required to divide a router into 2 routers	Yes, for clients who ask for it	Implement vrf in specific routers
Do technicians require ssh to connect to remote clients	Yes, it would be efficient	Implement SSH for secure connection to clients
Do you have a secure connection from ISP to ISP	No, we don't, we need one	Implement IPsec vpn for neighboring isp
Does your servers give away, Ip's dynamically	No I think dhcp is required	Implement dhcp servers

Do you have Nat to connect to the internet	No we don't have and our clients and companies need access	Implement Nat
Do you technicians have training to implement all these various protocols	They do have training but more is needed for these advanced protocols	Train the technicians on networking
Is monitoring of the network required	Yes, it is required	Implement networking monitoring methods in the topology

Functional requirements

No	Source	Requirement Description	Performance	Information	Economics	security	efficiency	services	priority
1	Research	Implementing border gateway protocol BGP	✓						H
2	Research	DHCP should be enabled for Ip addressing	✓	✓			✓	✓	H
3	Research	OSPF should be enabled for internal routing	✓						H
4	Research	Implement firewall		✓		✓			H
5	Research	Implement monitoring of network	✓	✓		✓			H
6	Research	Create vlans	✓				✓	✓	H
7	Research	Implement network redundancy	✓		✓		✓		H
8	Research	Training of horizon employees		✓				✓	M
9	Research	Implement Ip Sec vpn	✓					✓	H
10	Research	Implement MPLS for	✓		✓		✓	✓	H

		quality of service							
11	Research	Implement VRF							M
12	Research	Implement SSH for remote connections				✓	✓	✓	M
13	Research	Implement ACL for security	✓			✓			H
14	Research	Implement Mail server		✓		✓			H
15	Research	Implement failover links for ISP interfaces	✓		✓	✓			H

Nonfunctional requirements

No	Source	Requirement Description	Performance	Information	Economics	security	efficiency	services	Priory
1	Research	Implement disaster recovery plan	✓		✓	✓			M
2	Research	Provide support for IPv4-to-IPv6 transition		✓				✓	L
3	Research	Provide technical support services to customers		✓				✓	H
4	Research	Provide support for cloud technologies		✓	✓			✓	L
5	Research	Provide remote troubleshooting					✓	✓	M
6	Research	Horizon providing a website for clients		✓			✓	✓	M
7	Research	Horizon should follow a global ISP guideline		✓					M

8	Research	Horizon should have 99.9% network uptime	✓			✓	✓		H
9	Research	All client data should be encrypted				✓	✓		H
10	Research	The network should operate with no more than 5% packet loss	✓			✓	✓		H
11	Research	Horizon maintaining compatibly with global isp's	✓						H
12	Research	Horizons system should handle 50000 users simultaneously	✓				✓		H
13	Research	Horizons network should be easy configure for network engineers		✓					M
14	Research	Horizons network should be cost effective to maintain			✓				H
15	Research	Horizon infrastructure should minimize environmental impact		✓					L

Market Analysis:

Characteristics	Candidate 1: Horizon Technologies	Candidate 2: Mena Cloud	Candidate 3: yahsat Satellites
Benefits			

Brief description of the business benefits that would be realized for this candidate.	An internet service provider like horizon technologies provides internet access to local companies and government agencies which benefits with communication speeds and access to online resources	A cloud network provider like Mena cloud based in the middle east benefits companies with cost effective IT resources and enhanced operations	A satellite provider like Yahsat satellites provides access to remote areas not covered by normal networks which benefits with connectivity to underserved areas and regions
Servers and Workstations A description of the servers and workstations needed to support this candidate.	CPU: intel Xeon E5 or higher for an ISP Memory: 16 GB or higher. Storage: 1TB SSD or higher.	CPU: EC2, VPS. "Varies on instance type" Memory: customizable Storage: customizable	CPU: intel Xeon E5 Memory: 16 GB or higher. Storage: 500 GB HDD or higher.
Software Tools Needed Software tools needed to design and build the candidate (e. g., database management system, emulators, operating systems, languages, etc.). Not generally applicable if applications software packages are to be purchased.	EVE NG or GNS3 to design a realistic topology. IDS or IPS for monitoring and security	Cloud management console DB management System Development tools, for developing applications	Satellite communication software. Data analytic tools
Application Software A description of the software to be purchased, built, accessed, or some combination of these techniques.	SolarWinds Network Performance Monitor	VMware Microsoft Hyper-V	Yahsat Ground Segment Manager SolarWinds Network Performance Monitor
Method of Data Processing Generally some combination of: on-line, batch, deferred batch, remote batch, and real-time.	Real-time processing Batch processing	Real-time	Real-time
Output Devices and Implications	Routers, switches...	Web browser	Satellite receiver

A description of output devices that would be used, special output requirements, (e.g. network, preprinted forms, etc.), and output considerations (e.g., timing constraints).			
Input Devices and Implications A description of Input methods to be used, input devices (e.g., keyboard, mouse, etc.), special input requirements, (e.g. new or revised forms from which data would be input), and input considerations (e.g., timing of actual inputs).	Keyboard, Mouse,	Keyboard, Mouse.	Keyboard, Mouse,
Storage Devices and Implications Brief description of what data would be stored, what data would be accessed from existing stores, what storage media would be used, how much storage capacity would be needed, and how data would be organized.	Store network configuration data, traffic logs, performance metrics, and customer information locally on servers	Store user data, application data, configuration data, and logs on cloud-based storage infrastructure.	Store network configuration data, traffic logs, performance metrics, and customer information locally on servers

Implementation



The figures below show the internal ospf and external BGP routing with both protocols being redistributed into each other, R4 is the border router of Batelco ISP

```
R4
!
interface Serial1/2
  no ip address
  shutdown
  serial restart-delay 0
!
interface Serial1/3
  no ip address
  shutdown
  serial restart-delay 0
!
router ospf 10
  redistribute bgp 100 subnets
  network 10.0.0.4 0.0.0.3 area 0
  network 10.0.0.8 0.0.0.3 area 0
!
router bgp 100
  bgp log-neighbor-changes
  network 30.0.0.0 mask 255.255.255.252
  redistribute ospf 10
  neighbor 30.0.0.1 remote-as 200
!
ip forward-protocol nd
!
```

R3 of Batelco ISP show only ospf connection

```
!
interface Ethernet0/0
  ip address 10.0.0.10 255.255.255.252
!
interface Ethernet0/1
  ip address 10.0.0.13 255.255.255.252
!
interface Ethernet0/2
  no ip address
  shutdown
!
interface Ethernet0/3
  no ip address
  shutdown
!
router ospf 10
  network 10.0.0.8 0.0.0.3 area 0
  network 10.0.0.12 0.0.0.3 area 0
!
ip forward-protocol nd
!
!
```



the figure below will show the internal routing of ospf and external of BGP but for STC isp, R8 is the border router of the STC network.

```
R8
interface Serial11/3
 no ip address
 shutdown
 serial restart-delay 0
!
router ospf 20
 redistribute bgp 300 subnets
 network 20.0.0.0 0.0.0.3 area 0
 network 20.0.0.8 0.0.0.3 area 0
!
router bgp 300
 bgp log-neighbor-changes
 network 30.0.0.4 mask 255.255.255.252
 redistribute ospf 20
 neighbor 30.0.0.5 remote-as 200
!
ip forward-protocol nd
!
!
no ip http server
no ip http secure-server
!
!
--More--
```

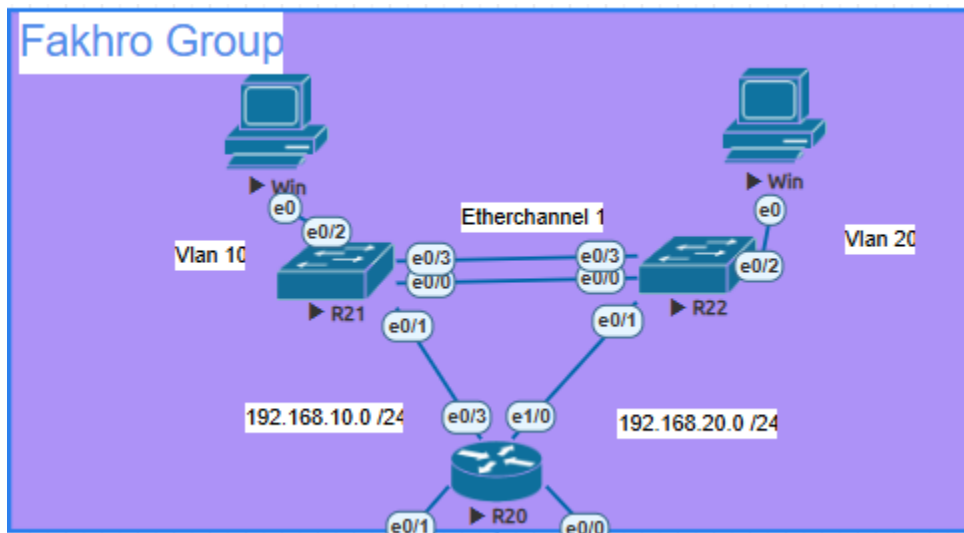
and finally in terms of internal and external routing here is the R7 which is the horizon border router connecting horizon technologies to both Batelco and STC it has the similar and ospf internal configuration and bgp external routing using different as numbers of course.

```

!
interface Serial2/3
  no ip address
  shutdown
  serial restart-delay 0
!
router bgp 200
  bgp log-neighbor-changes
  network 30.0.0.0 mask 255.255.255.252
  network 30.0.0.4 mask 255.255.255.252
  network 30.0.0.8 mask 255.255.255.252
  neighbor 30.0.0.2 remote-as 100
  neighbor 30.0.0.6 remote-as 300
  neighbor 30.0.0.10 remote-as 400
!
ip forward-protocol nd
!
!
no ip http server
no ip http secure-server
!
!
!
--More--

```

The configurations below will be the protocols requested by the Fakhro Group





the figure below shows the configuration for r20 which has inter vlan configured on 3.10 and 0.20 so that the pc's can go through their respective vlan into the router ospf and outside network.

```
R20
Ethernet0/3.10      192.168.10.1    YES NVRAM  up        up
Ethernet1/0         unassigned    YES NVRAM  up        up
Ethernet1/0.20      192.168.20.1    YES NVRAM  up        up
Ethernet1/1         unassigned    YES NVRAM  administratively down down
Ethernet1/2         unassigned    YES NVRAM  administratively down down
Ethernet1/3         unassigned    YES NVRAM  administratively down down
Router(config)#
```

The below figure will show the EtherChannel between both switches r21 and r22 , what EtherChannel does is that it takes 2 links and combines them as one, so if both links have a speed of 1gbps it will combine them into 2 gbps it also covers redundancy as if one links fails the other link takes over.

```
R21
I - stand-alone s - suspended
H - Hot-standby (LACP only)
R - Layer3      S - Layer2
U - in use      f - failed to allocate aggregator

M - not in use, minimum links not met
u - unsuitable for bundling
w - waiting to be aggregated
d - default port

Number of channel-groups in use: 1
Number of aggregators:          1

Group  Port-channel  Protocol    Ports
-----+-----+-----+-----
1      Po1 (SU)        LACP        Et0/0 (P)  Et0/3 (P)
```



HORIZON

```
R21
Switch(config)#
Switch(config)#
Switch(config)#show etherchannel detail
      ^
% Invalid input detected at '^' marker.

Switch(config)#do show etherchannel detail
      Channel-group listing:
      -----

Group: 1
-----
Group state = L2
Ports: 2    Maxports = 4
Port-channels: 1 Max Port-channels = 4
Protocol:   LACP
Minimum Links: 0

      Ports in the group:
      -----
Port: Et0/0
-----
```

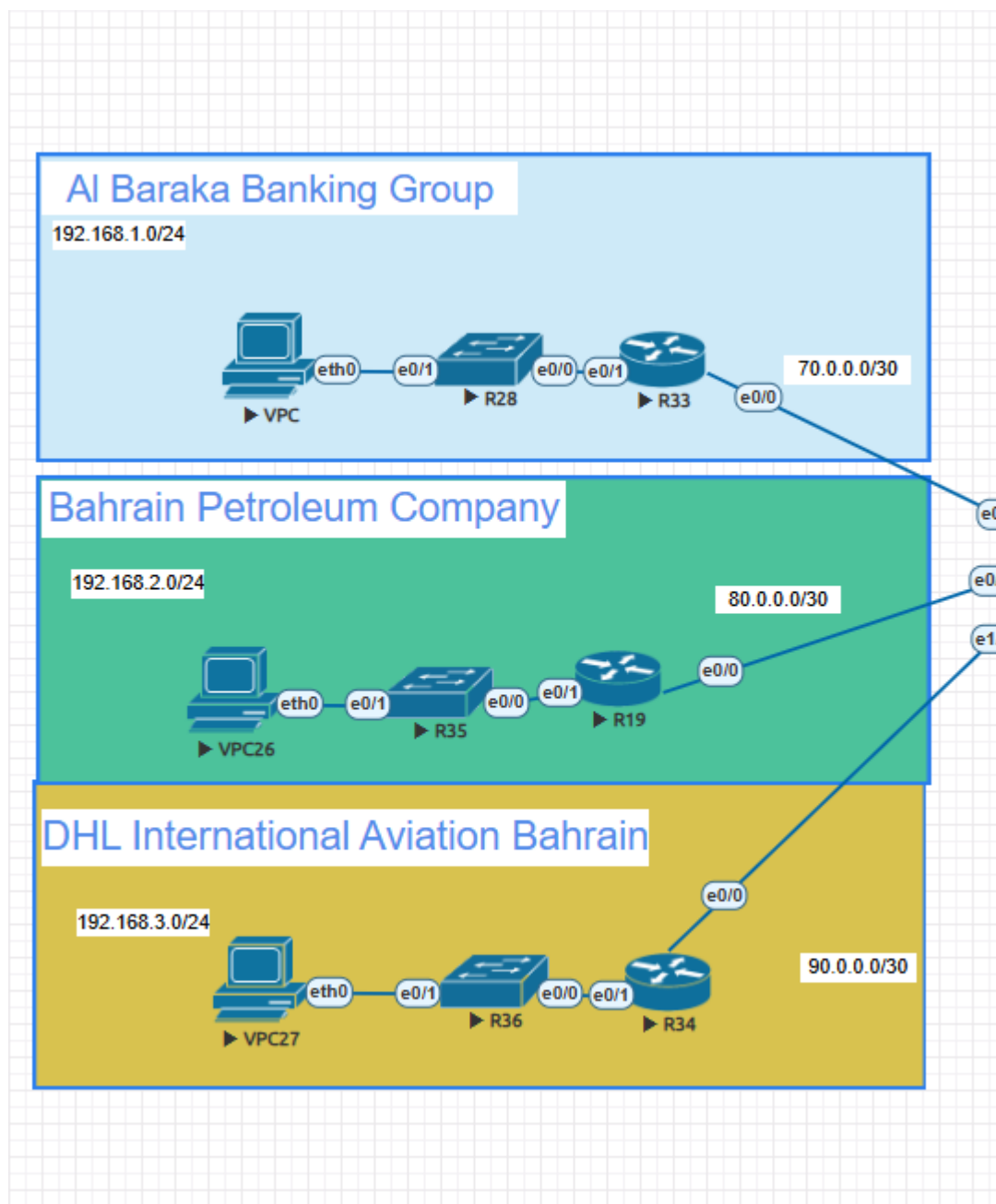
The figures below will show encapsulation mode of vlan interfaces and the vlan names on both r21 and r22



HORIZON

```
R21
switchport mode trunk
!
interface Ethernet0/0
switchport trunk encapsulation dot1q
switchport mode trunk
duplex auto
channel-group 1 mode active
!
interface Ethernet0/1
switchport trunk encapsulation dot1q
switchport mode trunk
duplex auto
!
interface Ethernet0/2
switchport access vlan 10
switchport mode access
duplex auto
!
interface Ethernet0/3
switchport trunk encapsulation dot1q
switchport mode trunk
duplex auto
channel-group 1 mode active
!
```

The configurations below will show r18 and respective company border routers of albaraka banking group branch 2, Bahrain petroleum company and DHL international aviation Bahrain.



the figures below will show the VRF configuration of r18 which is the horizon border router to the 3 companies, what VRF does is that it makes one router acts like 3 separate routers each with its own routing table, the reason is that to lower funds of buying more border routers for horizon, and it faster.



```
R18
Router>
Router>
Router>ena
Router#show vrf
  Name                Default RD        Protocols    Interfaces
  BLUE                 1:1              ipv4         Et0/0
                   Et0/1
                   Et0/2
                   Et0/3
                   Et1/0
Router#
```

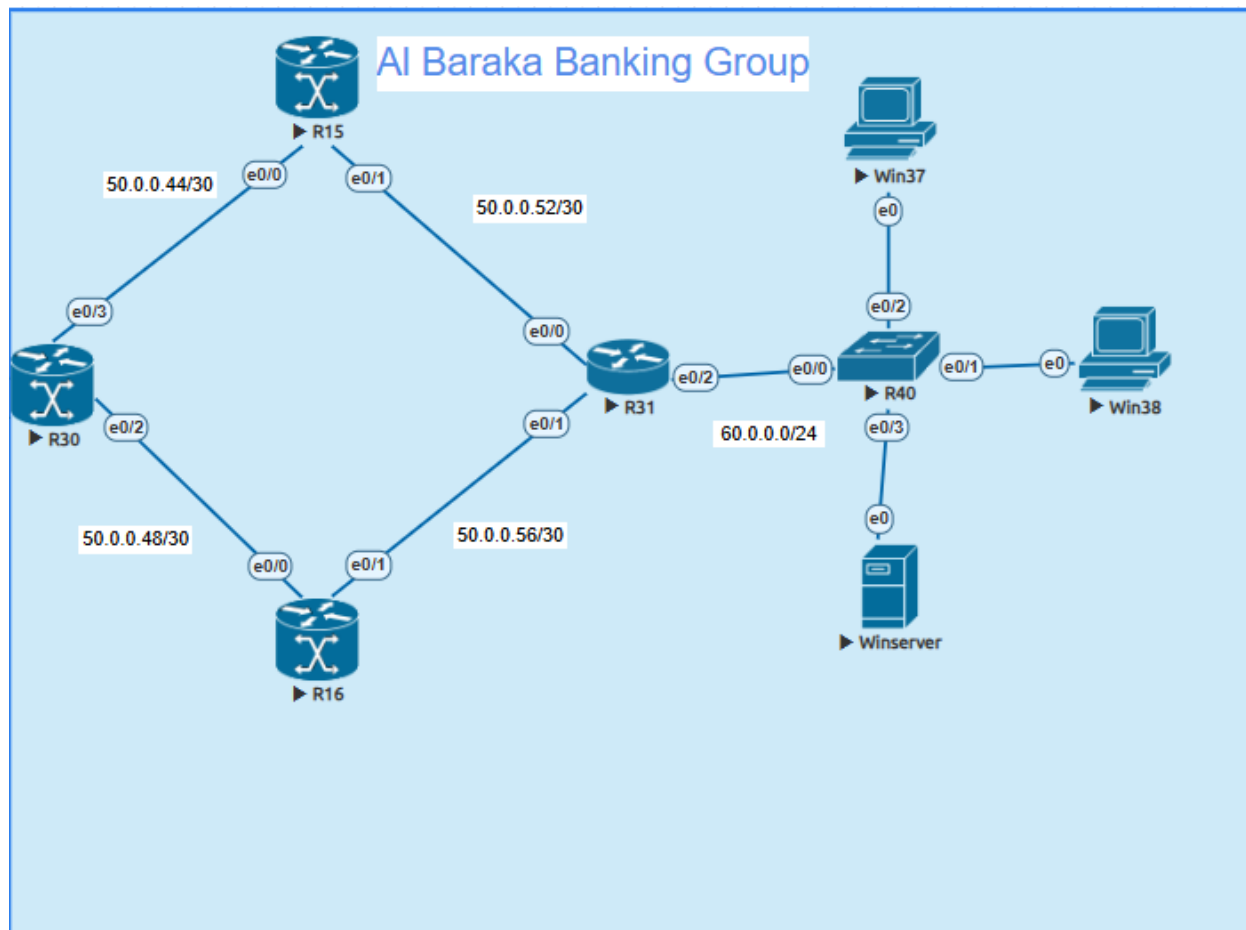
The figure below shows all the different interfaces that are linked with the rd1:1 vrf called BLUE and their Ip addresses, and you can that the protocol is up

```
Router#show ip vrf interface
Interface            IP-Address        VRF            Protocol
Et0/0                50.0.0.30         BLUE           up
Et0/1                50.0.0.26         BLUE           up
Et0/2                70.0.0.1          BLUE           up
Et0/3                80.0.0.1          BLUE           up
Et1/0                90.0.0.1          BLUE           up
Router#
```

the figure below will show the ACL configurations request by al baraka banking group branch 2, they requested the horizon close certain ports when they deemed to dangerous. The acl's were configured in router 18 and put into int e0/2 since it's the link connected to baraka branch 2

```
Extended IP access list 101
 10 deny ip host 192.168.1.5 192.168.2.0 0.0.0.255
 20 permit ip any any (913 matches)
Extended IP access list 102
 10 deny tcp 192.168.1.0 0.0.0.255 any eq www
 20 deny tcp 192.168.1.0 0.0.0.255 any eq 443
 30 deny tcp 192.168.1.0 0.0.0.255 any eq telnet
 40 deny tcp 192.168.1.0 0.0.0.255 any eq smtp
 50 permit ip any any
```

The configurations below will show the protocols requested by al baraka banking group main branch.



The figure below will show the QOS services that are provided by horizon for special customers who will request them proving they pay the extra fee, Al Baraka banking group requested MPLS which to simplify turns the routing table Ip addresses into labels to make packets travel faster.

R15

```
Router(config)#do show mpls interfaces
Interface      IP      Tunnel  BGP  Static  Operational
Ethernet0/0    Yes (ldp)  No      No   No      Yes
```

The below figures shows the labels of each network that is connected al baraka banking group



HORIZON

```
R15
Router(config)#do show mpls ldp bindings
lib entry: 3.3.3.0/24, rev 2
    local binding: label: imp-null
lib entry: 10.0.0.4/30, rev 52
    local binding: label: 38
lib entry: 10.0.0.8/30, rev 50
    local binding: label: 37
lib entry: 10.0.0.12/30, rev 58
    local binding: label: 41
lib entry: 20.0.0.0/30, rev 54
    local binding: label: 39
lib entry: 20.0.0.8/30, rev 56
    local binding: label: 40
lib entry: 30.0.0.0/30, rev 48
    local binding: label: 36
lib entry: 30.0.0.4/30, rev 46
    local binding: label: 35
lib entry: 30.0.0.8/30, rev 38
    local binding: label: 31
lib entry: 50.0.0.0/30, rev 32
    local binding: label: 28
lib entry: 50.0.0.4/30, rev 34
    local binding: label: 29
lib entry: 50.0.0.8/30, rev 30
```

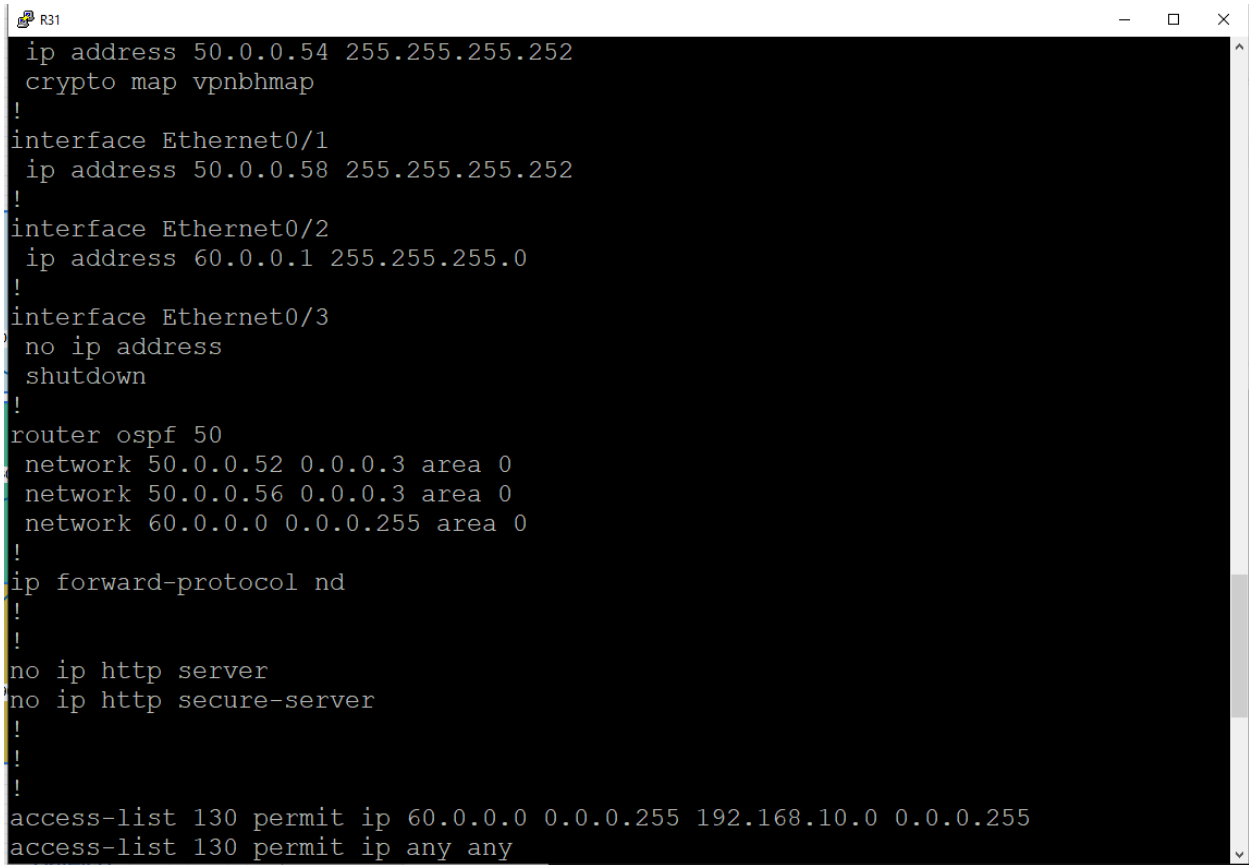
The below figures will show IPsec vpn tunnel requested by albakra banking group and fakhro group, what it is does is that it establishes a secure tunnel of communication between end devices making sure every packet that travels is encrypted for security reasons.

The figure is for r31 of albaraka, same configuration was done at r20 of fakhro group, just changing the peers and the interface for the crypto map



HORIZON

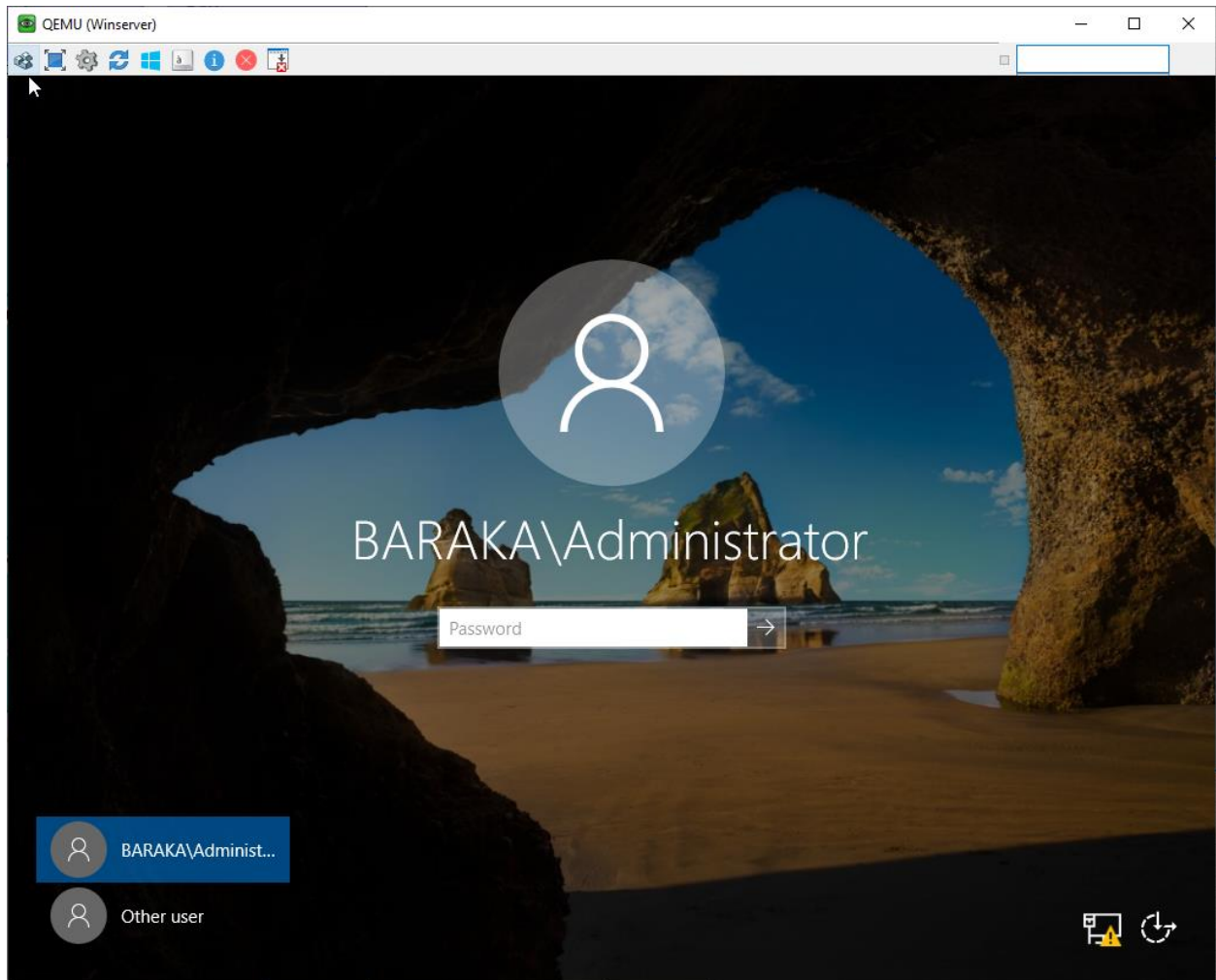
```
R31
crypto isakmp policy 10
  encr aes 256
  authentication pre-share
  group 5
crypto isakmp key vpnbhkey address 50.0.0.42
!
!
crypto ipsec transform-set vpnbhset esp-aes esp-sha-hmac
  mode tunnel
!
!
!
crypto map vpnbhmap 10 ipsec-isakmp
  description This VPN connects Branch office
  set peer 50.0.0.42
  set transform-set vpnbhset
  match address 130
!
!
!
!
!
interface Ethernet0/0
  ip address 50.0.0.54 255.255.255.252
```



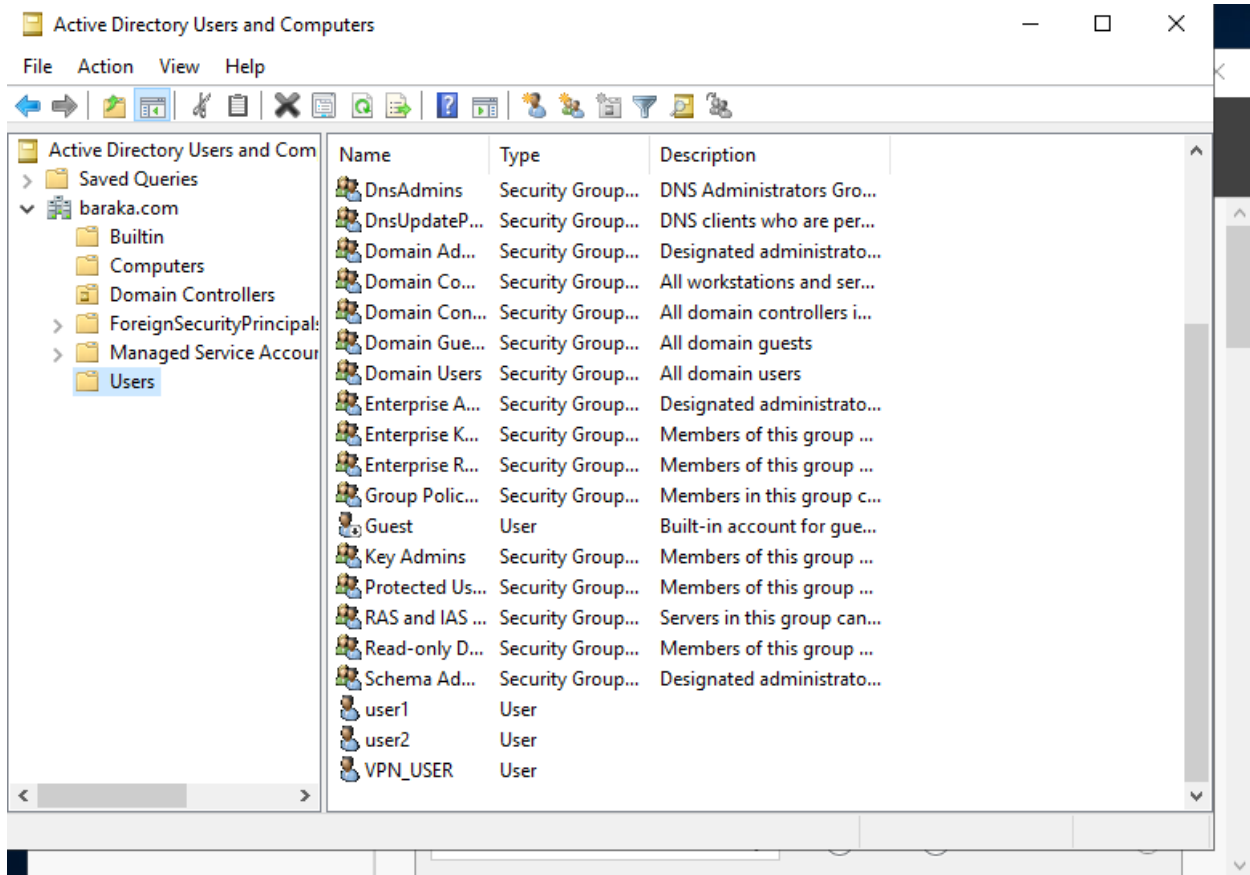
```
ip address 50.0.0.54 255.255.255.252
crypto map vpnbhmap
!
interface Ethernet0/1
ip address 50.0.0.58 255.255.255.252
!
interface Ethernet0/2
ip address 60.0.0.1 255.255.255.0
!
interface Ethernet0/3
no ip address
shutdown
!
router ospf 50
network 50.0.0.52 0.0.0.3 area 0
network 50.0.0.56 0.0.0.3 area 0
network 60.0.0.0 0.0.0.255 area 0
!
ip forward-protocol nd
!
!
no ip http server
no ip http secure-server
!
!
!
access-list 130 permit ip 60.0.0.0 0.0.0.255 192.168.10.0 0.0.0.255
access-list 130 permit ip any any
```

Albaraka banking group also request an installation of windows server, to create its own domain of baraka.com for its users, and set the win server as it dhcp server proving Ip addresses automatically for new devices connected to the domain.

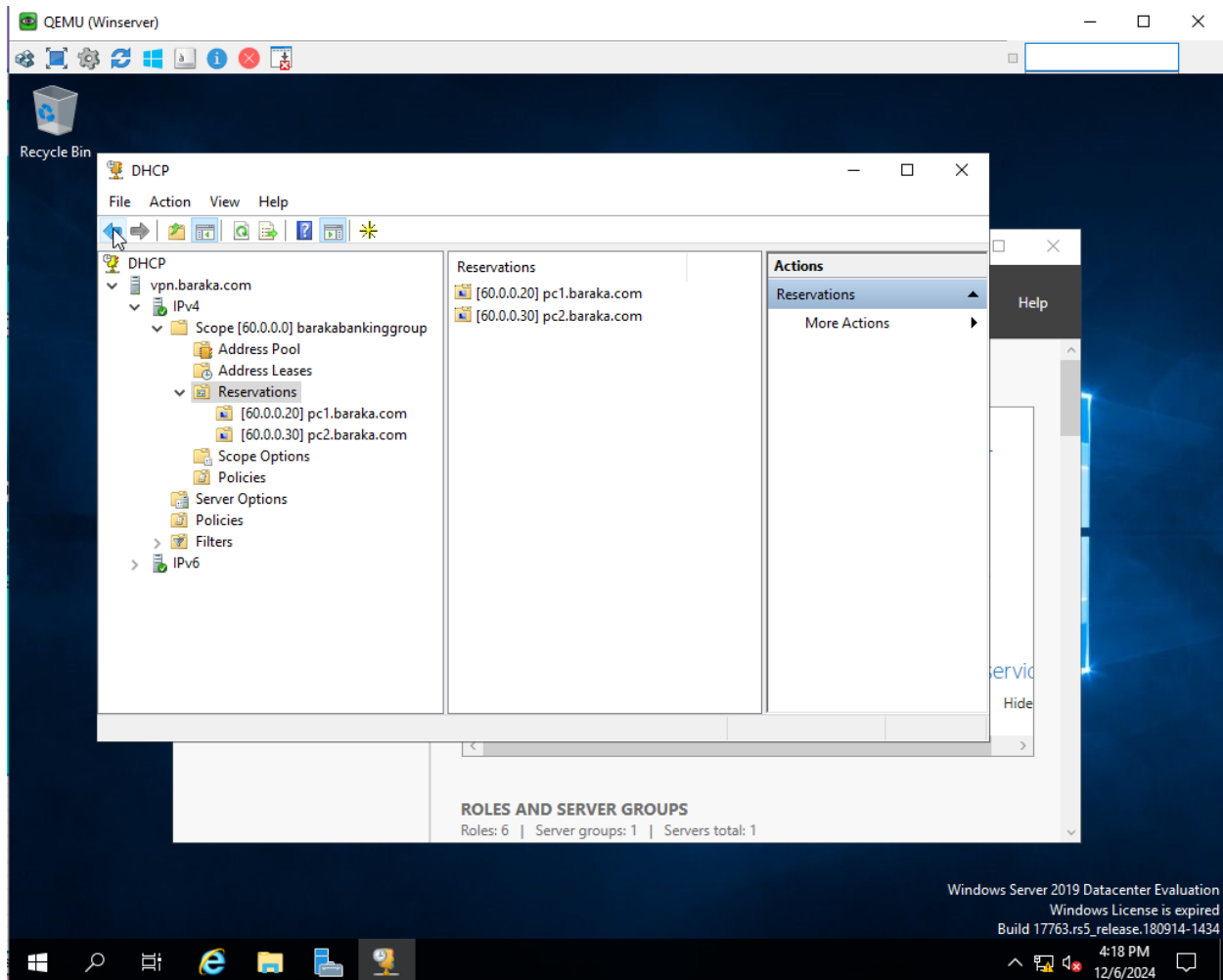
The figure below shows the administrative account for the windows server linked to the newly created domain

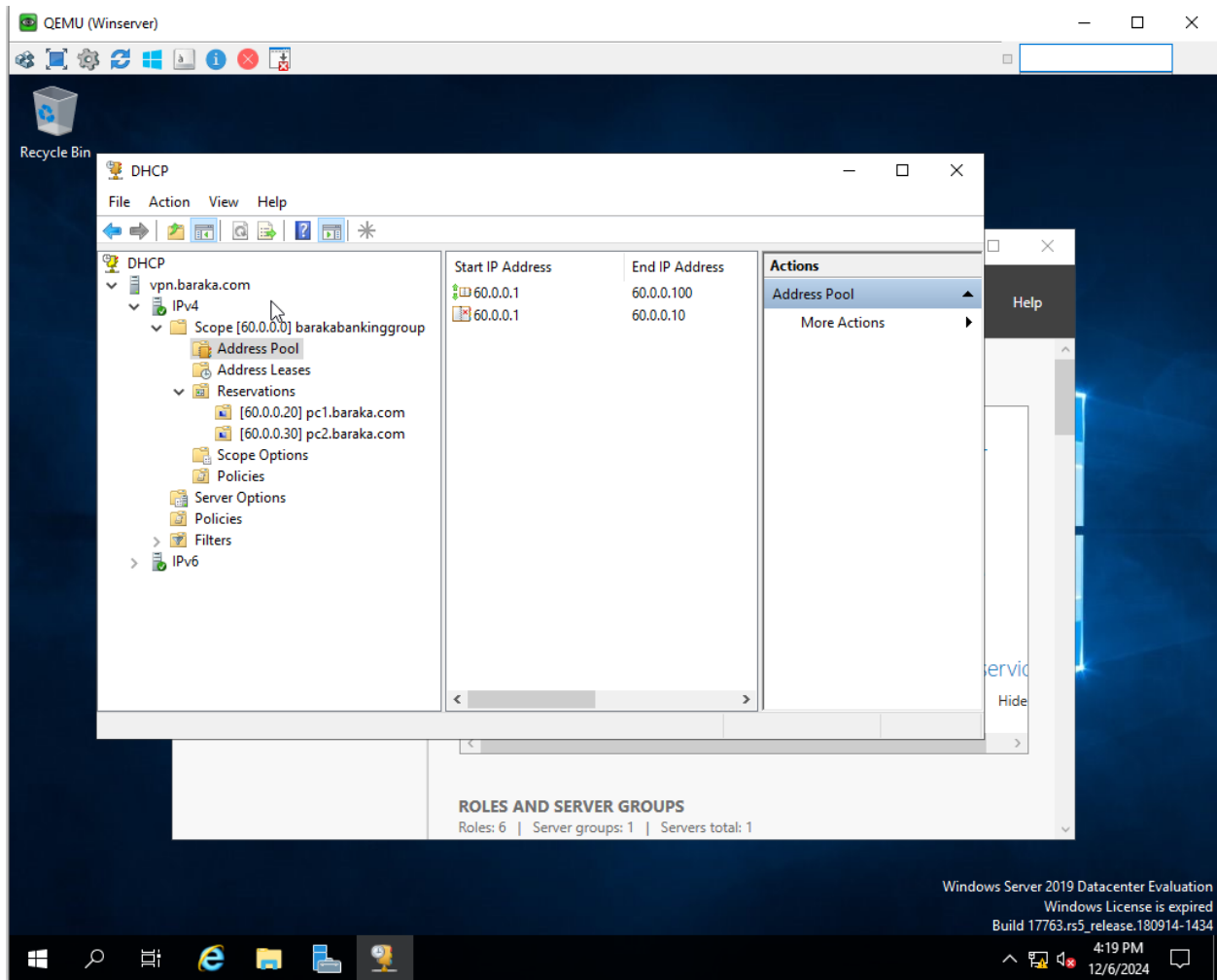


The figure below shows the domain create in active directory with three accounts created for its users after installing active directory tools in the win server.



The figure below shows the DHCP provided by the baraka windows server, it was made to give ip addresses ranged from 60.0.0.10 to 60.0.0.100 for newly added devices on albaraka network

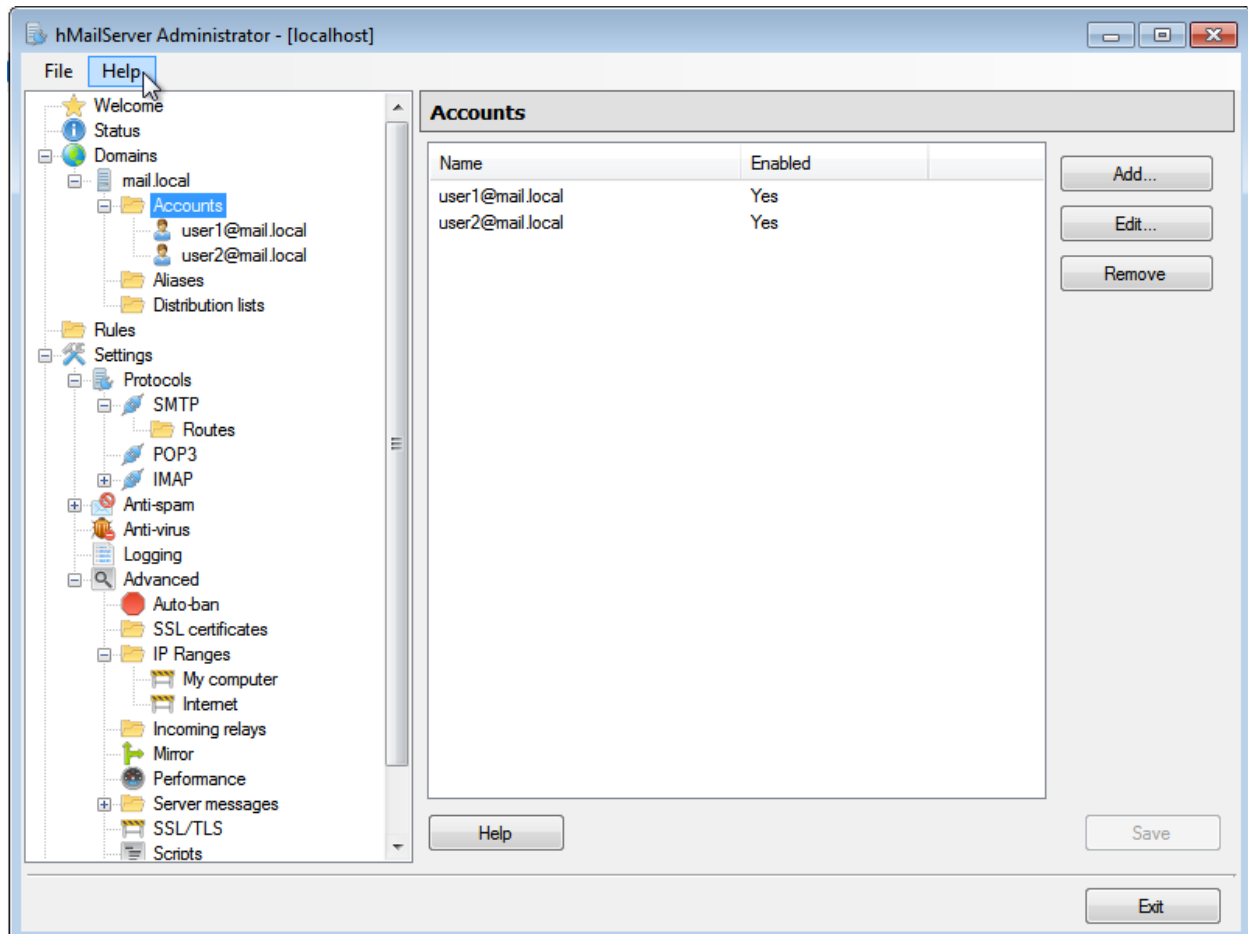




The figures below show the mail service requested by al baraka banking group main branch for its users, it was configured using thunderbird and hMail. As @mail. local, these are the created users for both pc using hmail



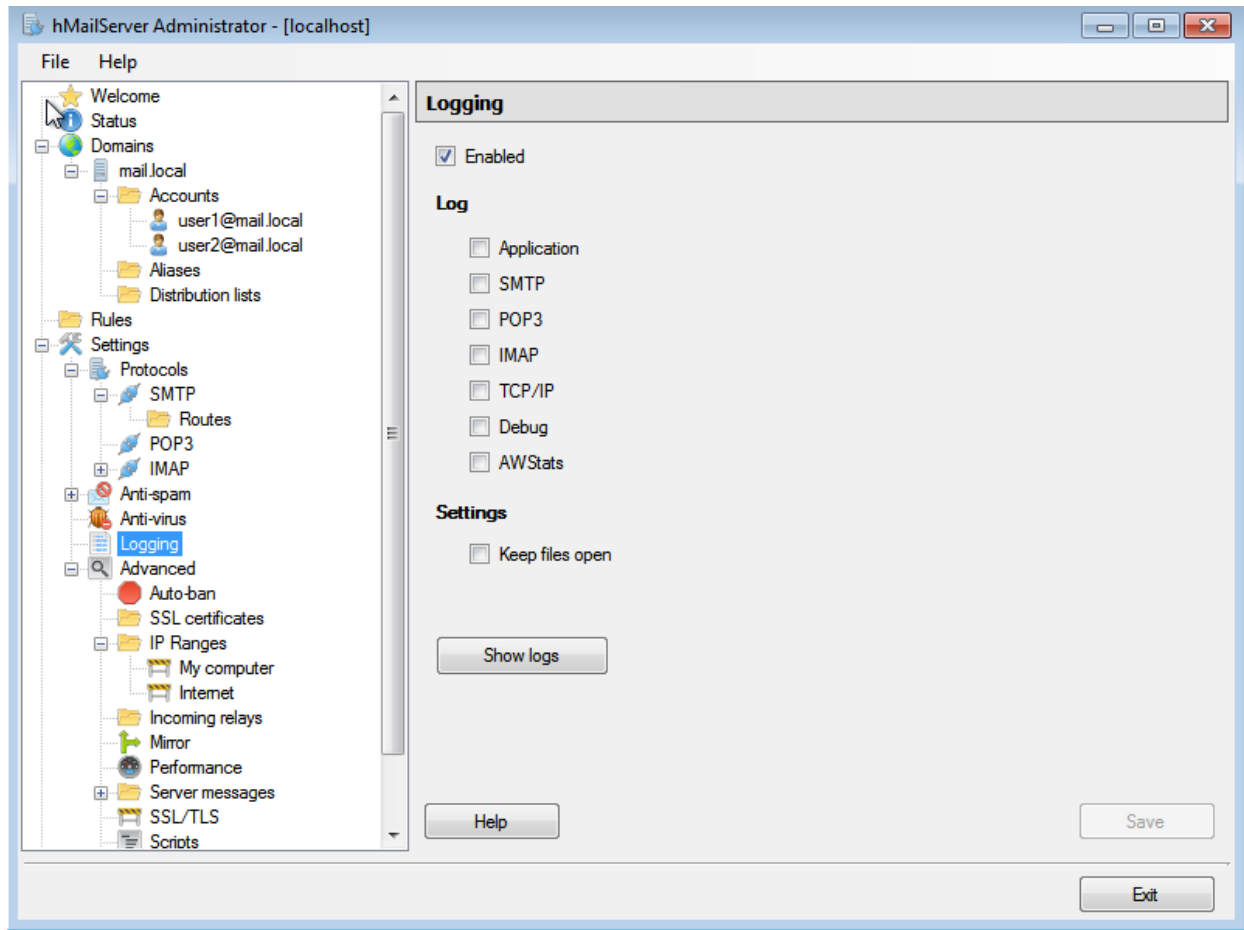
HORIZON



Logging was also enabled for security purposes



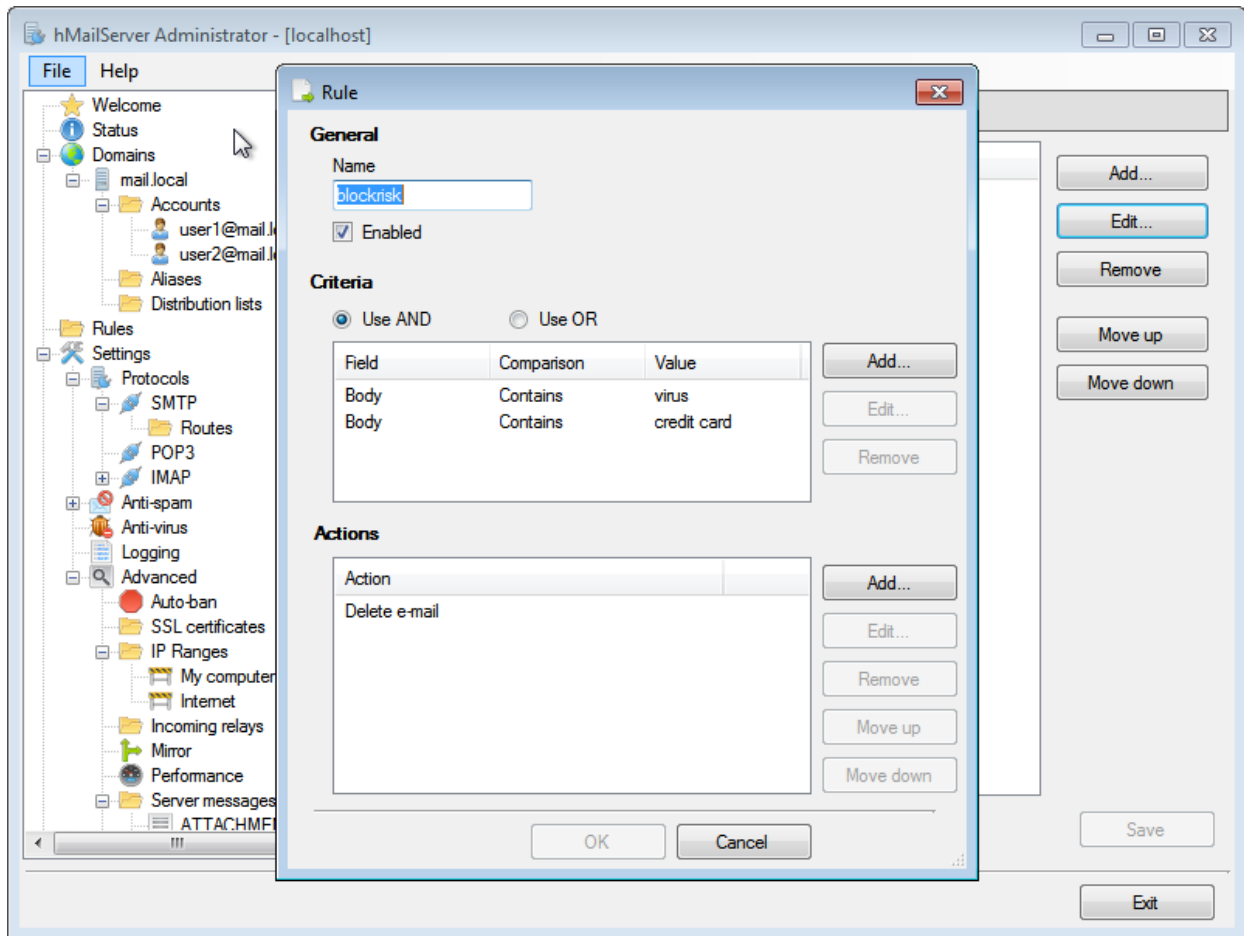
HORIZON



Another rule requested by al baraka banking group is that whenever a client receives an email that contains the word virus or credit card , the email should be deleted and no reach its destination, so that was also configured by horizon technologies



HORIZON



Testing

Both figures below will show that dhcp is working and running in the albaraka banking group main branch



HORIZON

```
C:\Windows\system32\cmd.exe

Ethernet adapter Local Area Connection:

    Connection-specific DNS Suffix  . : baraka.com
    Link-local IPv6 Address . . . . . : fe80::cda6:661b:91c3:14fa%11
    IPv4 Address. . . . . : 60.0.0.30
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 60.0.0.1

Tunnel adapter isatap.baraka.com:

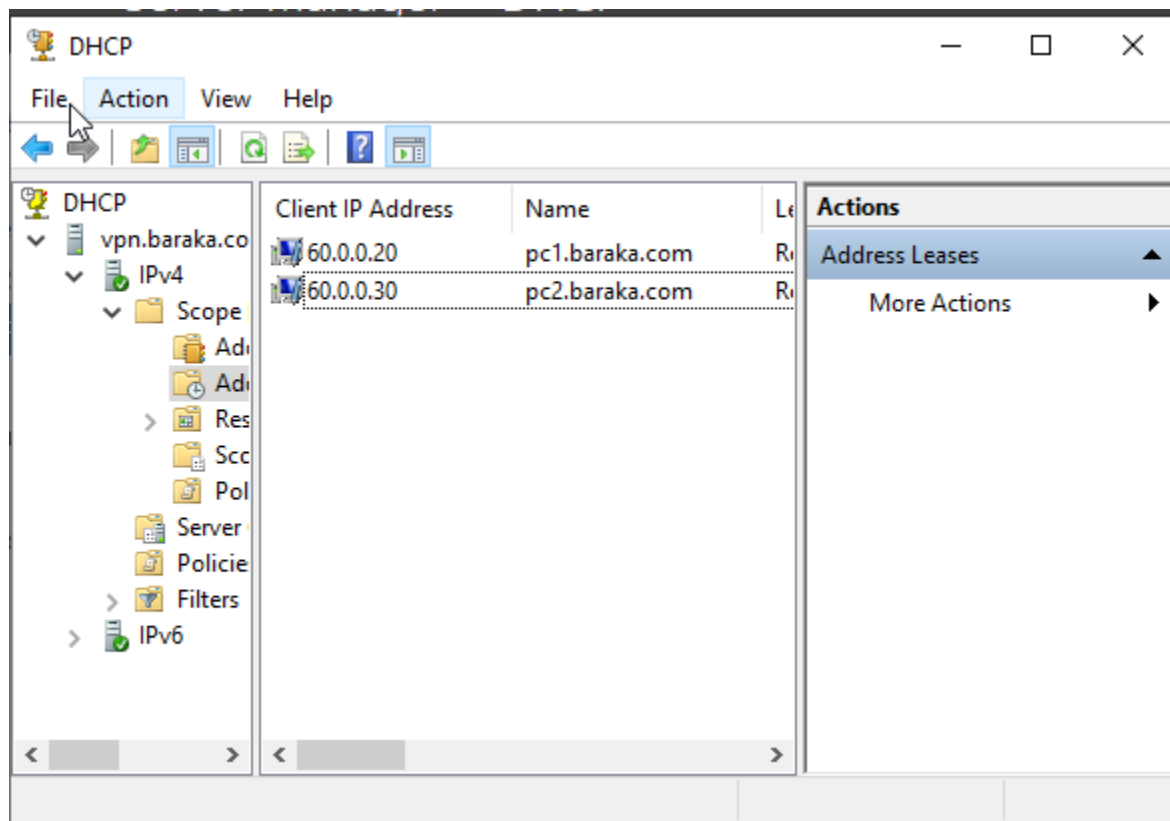
    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . : baraka.com

Tunnel adapter 6T04 Adapter:

    Connection-specific DNS Suffix  . : baraka.com
    IPv6 Address. . . . . : 2002:3c00:1e::3c00:1e
    Default Gateway . . . . . :

C:\Users\user2>
C:\Users\user2>
C:\Users\user2>
C:\Users\user2>
```

This figure shows that the win server acting like a dhcp server has two addresses one for pc1 and one for pc2 with their addresses



The figures below shows that active directory domain is working and that pc2 and pc1 are a part of albaraka network



View your basic network information and set up connections



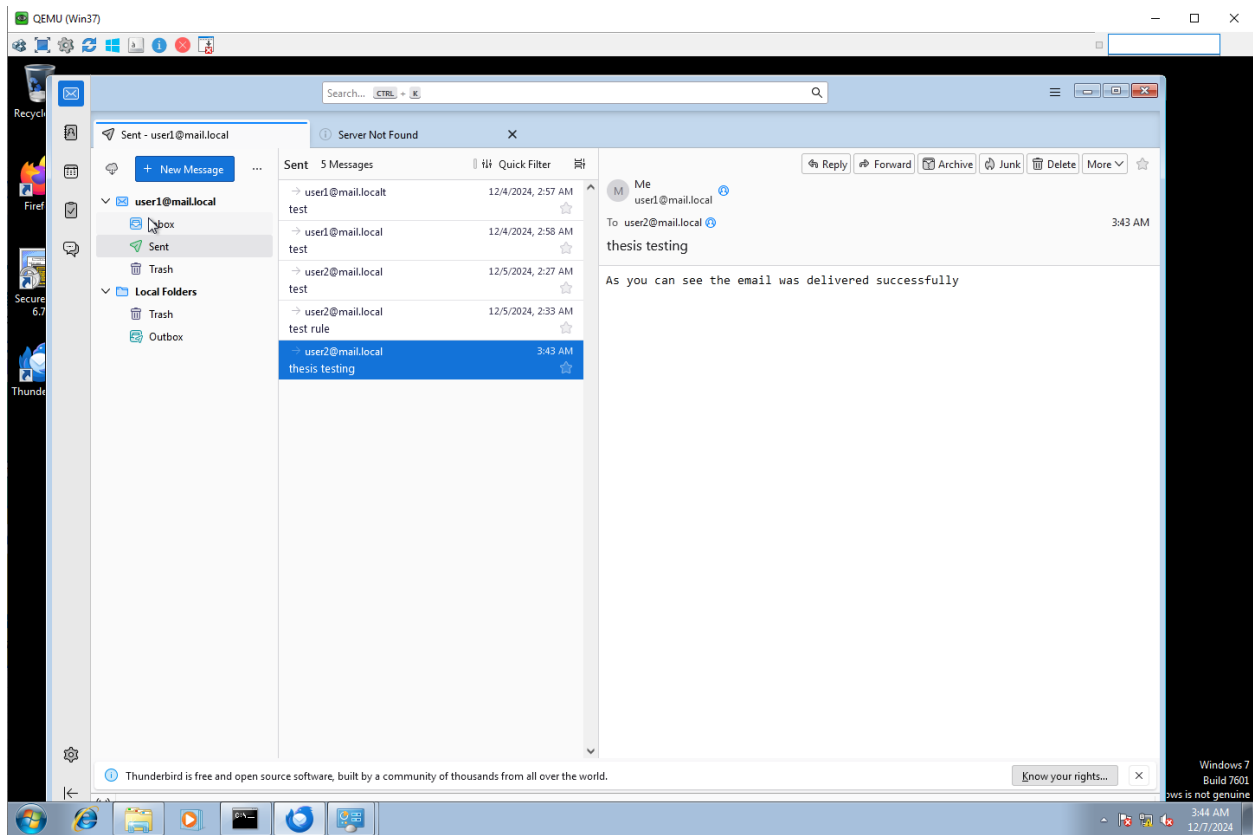
View your active networks [Connect or disconnect](#)

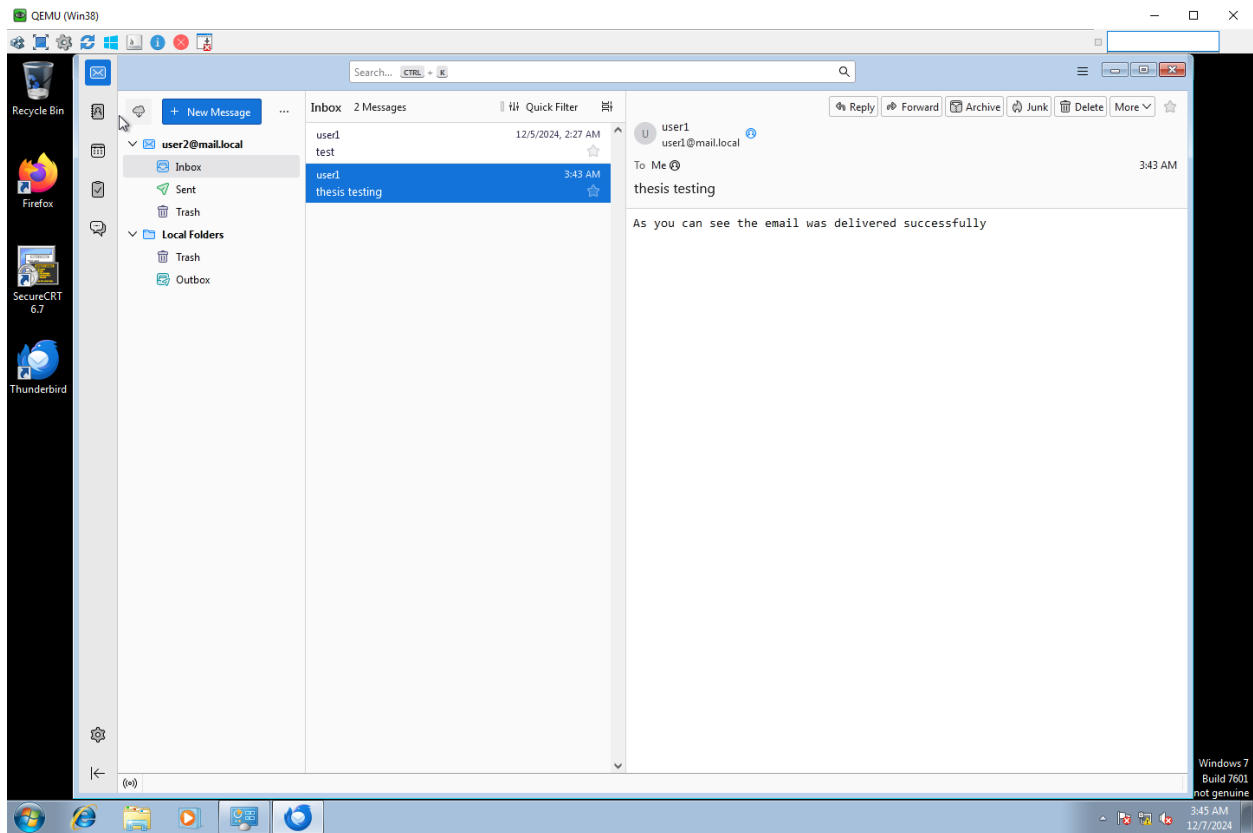
 **baraka.com**
Domain network

Access type: No Internet access
Connections:  Local Area Connection

[Change your networking settings](#)

The figures below will show that the mail server is successfully running sending a email from pc1 to pc2





The figure below will show that Ip sec tunnel is up and running successfully, you can see that the trace route is showing nothing because of the encryption.



HORIZON

```
R20
Router(config)#do tr
Router(config)#do traceroute 50.0.0.53
Type escape sequence to abort.
Tracing the route to 50.0.0.53
VRF info: (vrf in name/id, vrf out name/id)
 1  *  *  *
 2  *  *  *
 3  *  *  *
 4  *  *  *
 5  *  *  *
 6  *  *  *
 7  *  *  *
 8  *  *  *
 9  *  *  *
10  *  *  *
11  *  *  *
12  *  *  *
13  *  *  *
14  *  *  *
15  *  *  *
16  *  *  *
17  *  *  *
18  *  *  *
19  *  *  *
```

```
R20
Router(config)#do show crypto ipsec sa

interface: Ethernet0/2
  Crypto map tag: vpnbmap, local addr 50.0.0.42

  protected vrf: (none)
  local  ident (addr/mask/prot/port): (0.0.0.0/0.0.0.0/0/0)
  remote ident (addr/mask/prot/port): (0.0.0.0/0.0.0.0/0/0)
  current_peer 50.0.0.54 port 500
    PERMIT, flags={origin_is_acl,ipsec_sa request_sent}
    #pkts encaps: 0, #pkts encrypt: 0, #pkts digest: 0
    #pkts decaps: 0, #pkts decrypt: 0, #pkts verify: 0
    #pkts compressed: 0, #pkts decompressed: 0
    #pkts not compressed: 0, #pkts compr. failed: 0
    #pkts not decompressed: 0, #pkts decompress failed: 0
    #send errors 11, #recv errors 0

    local crypto endpt.: 50.0.0.42, remote crypto endpt.: 50.0.0.54
    plaintext mtu 1500, path mtu 1500, ip mtu 1500, ip mtu idb Ethernet0/2
    current outbound spi: 0x0(0)
    PFS (Y/N): N, DH group: none

  inbound esp sas:
```



The figure below shows that the MPLS in router 30 is working and showing the tunnel id and the next hop

```
R30
Router>
Router>
Router>
Router>
Router>
Router>
Router>
Router>ena
Router#
Router#show mpl
Router#show mpls fr
Router#show mpls fo
Router#show mpls forwarding-table
Local      Outgoing  Prefix      Bytes Label  Outgoing  Next Hop
Label      Label     or Tunnel Id Switched     interface
16         No Label  50.0.0.52/30 0           Et0/3       50.0.0.46
Router#
Router#
Router#
```

The figure below shows that BGP is running successfully

```
R7(config)#do show ip bgp
BGP table version is 113, local router ID is 30.0.0.9
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
               r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
               x best-external, a additional-path, c RIB-compressed,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

   Network          Next Hop        Metric LocPrf Weight Path
*> 10.0.0.4/30       30.0.0.2         0           0 100 ?
*> 10.0.0.8/30       30.0.0.2         0           0 100 ?
*> 20.0.0.0/30       30.0.0.6         0           0 300 ?
*> 20.0.0.8/30       30.0.0.6         0           0 300 ?
* 30.0.0.0/30        30.0.0.2         0           0 100 i
*>                   0.0.0.0         0          32768 i
* 30.0.0.4/30        30.0.0.6         0           0 300 i
*>                   0.0.0.0         0          32768 i
R7(config)#
```



The figure below shows that vrf is in R18 working by running show ip route command and getting nothing because we didn't specify the vrf name.

```
R18
Router(config)#
Router(config)#
Router(config)#
Router(config)#
Router(config)#
Router(config)#
Router(config)#do show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2
        i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
        ia - IS-IS inter area, * - candidate default, U - per-user static route
        o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
        a - application route
        + - replicated route, % - next hop override

Gateway of last resort is not set
```

And if we run the same command with VRF Blue at the end we get the real Ip routes of router 18

```
R18
Router(config)#do show ip route vrf BLUE

Routing Table: BLUE
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2
        i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
        ia - IS-IS inter area, * - candidate default, U - per-user static route
        o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
        a - application route
        + - replicated route, % - next hop override

Gateway of last resort is not set

      50.0.0.0/8 is variably subnetted, 4 subnets, 2 masks
C       50.0.0.24/30 is directly connected, Ethernet0/1
L       50.0.0.26/32 is directly connected, Ethernet0/1
C       50.0.0.28/30 is directly connected, Ethernet0/0
L       50.0.0.30/32 is directly connected, Ethernet0/0
      70.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       70.0.0.0/30 is directly connected, Ethernet0/2
L       70.0.0.1/32 is directly connected, Ethernet0/2
      80.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
```



HORIZON

The figure below shows that ACL security is running for albarka banking branch 2 at 192.16.1.0 for different ports

 R18

```
20 permit ip any any (215 matches)
Extended IP access list 102
10 deny tcp 192.168.1.0 0.0.0.255 any eq www
20 deny tcp 192.168.1.0 0.0.0.255 any eq 443
30 deny tcp 192.168.1.0 0.0.0.255 any eq telnet
40 deny tcp 192.168.1.0 0.0.0.255 any eq smtp
50 permit ip any any
```

The figure below shows Ip routes of r7 which is the border router of horizon and shows the Ip addresses, it can ping all the Ip addresses in the topology which is proof that all internal and external routing protocols are working.

```
R7
Gateway of last resort is not set

10.0.0.0/8 is variably subnetted, 5 subnets, 2 masks
B    10.0.0.0/30 [20/20] via 30.0.0.2, 00:01:22
B    10.0.0.4/30 [20/0] via 30.0.0.2, 00:01:52
B    10.0.0.8/30 [20/0] via 30.0.0.2, 00:01:52
B    10.0.0.12/30 [20/20] via 30.0.0.2, 00:01:22
B    10.10.10.0/24 [20/30] via 30.0.0.2, 00:00:20
20.0.0.0/8 is variably subnetted, 5 subnets, 2 masks
B    20.0.0.0/30 [20/0] via 30.0.0.6, 00:01:50
B    20.0.0.4/30 [20/20] via 30.0.0.6, 00:01:20
B    20.0.0.8/30 [20/0] via 30.0.0.6, 00:01:50
B    20.0.0.12/30 [20/20] via 30.0.0.6, 00:01:20
B    20.20.20.0/24 [20/30] via 30.0.0.6, 00:00:18
30.0.0.0/8 is variably subnetted, 6 subnets, 2 masks
C    30.0.0.0/30 is directly connected, Serial1/0
L    30.0.0.1/32 is directly connected, Serial1/0
C    30.0.0.4/30 is directly connected, Serial1/2
L    30.0.0.5/32 is directly connected, Serial1/2
C    30.0.0.8/30 is directly connected, Serial1/3
L    30.0.0.9/32 is directly connected, Serial1/3
50.0.0.0/30 is subnetted, 14 subnets
B    50.0.0.0 [20/0] via 30.0.0.10, 00:00:20
B    50.0.0.4 [20/20] via 30.0.0.10, 00:00:20
B    50.0.0.8 [20/30] via 30.0.0.10, 00:00:20
B    50.0.0.12 [20/40] via 30.0.0.10, 00:00:20
B    50.0.0.16 [20/50] via 30.0.0.10, 00:00:20
B    50.0.0.20 [20/0] via 30.0.0.10, 00:00:20
B    50.0.0.24 [20/60] via 30.0.0.10, 00:00:20
B    50.0.0.28 [20/50] via 30.0.0.10, 00:00:20
B    50.0.0.32 [20/20] via 30.0.0.10, 00:00:20
B    50.0.0.36 [20/30] via 30.0.0.10, 00:00:20
B    50.0.0.40 [20/40] via 30.0.0.10, 00:00:20
B    50.0.0.44 [20/30] via 30.0.0.10, 00:00:20
B    50.0.0.48 [20/30] via 30.0.0.10, 00:00:20
```

The figure below shows that intervlan routing is working and vlan 10 can ping vlan 20.

```
Administrator: C:\Windows\system32\cmd.exe

Connection-specific DNS Suffix  : 
Link-local IPv6 Address . . . . : fe80::c525:d3dc:6dfb:fed2%11
IPv4 Address. . . . . : 192.168.10.2
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . : 192.168.10.1

Tunnel adapter isatap.{E224E83C-90A8-45B5-B805-81B998611BD6}:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix  : 

Tunnel adapter 6T04 Adapter:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix  : 

C:\Users\Administrator>
C:\Users\Administrator>
C:\Users\Administrator>
C:\Users\Administrator>
C:\Users\Administrator>ping 192.168.20.1

Pinging 192.168.20.1 with 32 bytes of data:
Reply from 192.168.20.1: bytes=32 time<1ms TTL=255
Reply from 192.168.20.1: bytes=32 time<1ms TTL=255
Reply from 192.168.20.1: bytes=32 time<1ms TTL=255
Reply from 192.168.20.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.20.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

The figure below shows that VTP is working and that r22(switch) is a vtp client



```
R22
Switch(config)#do show vtp status
VTP Version capable      : 1 to 3
VTP version running      : 3
VTP Domain Name          : ABC
VTP Pruning Mode         : Disabled
VTP Traps Generation     : Disabled
Device ID                : aabb.cc01.6000

Feature VLAN:
-----
VTP Operating Mode       : Client
Number of existing VLANs : 6
Number of existing extended VLANs : 0
Maximum VLANs supported locally : 4096
Configuration Revision   : 0
Primary ID               : 0000.0000.0000
Primary Description      :
MD5 digest               : 0x00 0x00 0x00 0x00 0x00 0x00 0x00 0x00
                        : 0x00 0x00 0x00 0x00 0x00 0x00 0x00 0x00

Feature MST:
-----
```

The figure below shows R33 of Albaraka second branch pinging through vrf since normal ping wont work for VRF routers

```
R33
Router>
Router>
Router>ena
Router#ping vrf BLUE 50.0.0.54
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 50.0.0.54, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/2/5 ms
Router#
```



Test plan

Horizon Technologies' ISP project's testing phase comprised a thorough set of assessments to guarantee the network infrastructure's security, dependability, and performance for both the company and clients'. Ping tests were utilized in the first round of testing to verify that all devices—internal and external—were able to communicate with each other without any problems. Proper configurations and functional links throughout the infrastructure were guaranteed by this fundamental stage.

The setup and routing capabilities of the Border Gateway Protocol (BGP) were then verified through testing. In order to ensure precise and effective inter-ISP communication redistribution, this required transmitting data between users from one ISP to another. Route tracing was done to verify the encryption and safe data transfer between endpoints for the clients as part of a comprehensive assessment of the IPsec installation.

By adding other devices, such a router, a PC, and a Windows server, the DHCP server's operation was further evaluated in order to confirm that IP addresses were assigned automatically. Virtual Routing and Forwarding (VRF) and Multiprotocol Label Switching (MPLS) were thoroughly tested to ensure that packets were properly forwarded across MPLS routers and that routing tables were successfully segmented for various clients sharing a single router. Triple AAA was also tested on the Windows server for clients belonging to the Baraka Banking Group that had their own domain.

Acceptance test process and results

The results of Horizon Technologies' ISP network's acceptance testing phase were very positive, showing that the system fulfilled all predetermined specifications and operated as planned in a variety of scenarios. Every test confirmed that the system could provide secure communication, dependable routing, smooth connectivity, and effective network administration.

Strong communication connections throughout the infrastructure were confirmed by the successful completion of the connectivity tests, which involved pinging devices across the network ranging from local networks to other isp client's. This achievement confirmed that the basic architecture of the network was set up appropriately and operated dependably under typical circumstances in bahrain.

Tests of BGP routing were equally successful, showing that consumers from various ISPs could transfer data with ease. The project's objectives for inter-ISP connectivity were met by the seamless and effective data transfers made possible by the proper setup of routing tables and cross-ISP communication protocols.



Secure data transfer and encryption were validated by security tests that focused on IPsec capabilities. All communication channels passed the encryption validation tests, confirming the dependability of the established protocols and guaranteeing security to the clients

When it came to dynamically allocating IP addresses to newly connected devices, the DHCP server tests showed faultless operation. The efficiency of the server in controlling address distribution throughout the network was validated by the timely and accurate IP addresses that each device received.

To confirm packet traversal and routing table separation, MPLS and VRF implementations underwent extensive testing. The segmentation offered by VRF guaranteed unique and secure routing for various customers reducing load on horizon and speed for customers, and all packets were successfully routed through MPLS routers with the proper labels.

Overall, the results of the acceptance tests confirmed that the system met and exceeded expectations in terms of reliability, functionality, and security.

Discussion

System functionality

In order to provide internet access, Internet Service Providers (ISPs) must have a strong infrastructure of servers, routers, and switches to guarantee effective data flow. An ISP system's capacity to ensure dependable service, secure data transfer throughout its network, and offer seamless connectivity is what makes it functional and horizon is no different.

The goal of the Horizon Technologies project was to optimise important network components and apply cutting-edge protocols to improve the system's functionality. While IPsec was used to provide encrypted data transmission security, the project's top priority was setting up effective routing using Border Gateway Protocol (BGP) for inter-ISP communication. A DHCP server enables dynamic IP address management, guaranteeing dependable and automatic address distribution to linked devices. Multiprotocol Label Switching (MPLS) improved Quality of Service (QoS) by enabling effective packet labelling and routing. Furthermore, network segmentation was made possible using Virtual Routing and Forwarding (VRF) in order to separate and safeguard various client networks.

Performance and dependability were confirmed by extensive testing of the system's functionality. While packet traversal tests evaluated the functioning of MPLS and BGP routing, connectivity testing made sure that devices communicated with one other without any problems. The DHCP server was verified for dynamic address allocation, and security protocols such as IPsec were tested to verify encrypted data delivery. To provide efficient network isolation and segmentation, the VRF implementation was carefully examined.



All things considered, the project produced a very safe and useful system that satisfied the needs of Horizon Technologies' network and made sure it was dependable, scalable, and able to handle future expansion while retaining strong performance.

Archived objectives

The project's main goals of creating reliable connectivity, putting in place solid security measures, and guaranteeing network resilience were all accomplished. Virtual Routing and Forwarding (VRF) and dynamic routing protocols were used to segment and optimise internal communication in order to achieve internal connection. Border Gateway Protocol (BGP) was used to guarantee external connectivity since it is more optimal than ospf in long range networks, enabling smooth and effective communication between Horizon Technologies' border routers and other ISPs. ACLs were set up to filter harmful packets, VRF was used for routing privacy, and IPsec tunnels were used for encrypted data transmission

By creating several linkages between internal paths and ISP border routers, redundancy was added to the system, allowing for dynamic rerouting in the case of device or link failures. Continuous operations and reduced downtime are guaranteed by this architecture. Furthermore, the use of Multiprotocol Label Switching (MPLS), which is backed by QoS setups to prioritise important data, improved packet flow consistency and quality. By integrating a DHCP server to automate IP address distribution, scalability was ensured and the network could effectively handle expansion.

Objective	Status	Description
Ensure proper internal connectivity protocols.	Achieved	VRF was used to segment internal routing for some companies , optimizing privacy and communication.
Ensure external connectivity between ISPs.	Achieved	BGP was used for routing between ISP border routers, ensuring stable and scalable communication.
Enhance network security.	Achieved	IPsec, ACLs, were applied for data protection and threat mitigation.
Provide redundancy for reliability.	Achieved	Multiple links were established to enable dynamic rerouting during failures.
Optimize packet flow with MPLS.	Achieved	Provide faster communication for special customers using MPLS
Implement Mail Server	Achieved	Provide mail services for special customers

Automate IP address allocation.	Achieved	A DHCP server was configured for automated and scalable IP address management.
---------------------------------	----------	--

Project issues

Horizon Technologies' ISP project faced a number of obstacles throughout execution, necessitating flexibility and problem-solving skills. The biggest problem occurred when the EVE-NG image became corrupted after the network connections were finished. Progress was lost as a result, and the software had to be completely reinstalled. The installation of the Windows 10 and Windows Server images on eve ng, which might have overwhelmed the machine, was the suspected cause. Only the Windows Server image was deployed during the reconfiguration phase to prevent a repeat, enabling the project to continue uninterrupted.

The setup of VRF presented another difficulty for the Horizon project; at first, it was unable to connect to external networks. Many testing and troubleshooting were necessary to fix this problem. Furthermore, the project's broad topology resulted in lag while several routers were operating because of the system it was being run on, which prolonged the time needed to verify device connectivity and performance.

Relearning fundamental protocols, especially layer 2 protocols, which had been covered in previous coursework but were not thoroughly recalled, was another problem. To make sure everything was working it was necessary to go back to old material again.

These difficulties highlighted the significance of testing, effective resource management and time management, and preserving a solid foundation in networking principles.

Issues and limitations	Resolved method
EVE-NG image corrupted, requiring full reinstallation.	Reinstalled EVE-NG and avoided using multiple OS images, opting only for Windows Server.
VRF was not connecting to external networks.	Troubleshoot VRF configurations and resolved through iterative testing.
Lag caused by activating all routers in a large topology.	Tested devices individually to ensure connectivity without overloading the system.
Forgot foundational layer 2 protocols from earlier studies.	Revisited and relearned concepts from previous coursework to correctly implement switching protocols.



Backup plan

For Horizon Technologies' ISP infrastructure to be resilient and stay a long term, a solid backup strategy is necessary. Regular backups of IP allocations, router and switch configurations with all their protocols used, and a comprehensive topology file are safely kept in physical copies

Configuration backups for routing tables, protocol configurations, and security features like ACLs and VRF setups are kept separately in a separate physical location for increased security. By facilitating speedy recovery in the case of hardware malfunctions, misconfigurations, or cyberattacks, these files reduce downtime. Additionally, regular snapshots of the EVE-NG virtual environment image that we were given ensure that the network may be fully restored, safeguarding both device configurations and the overall project setup. Even while Eve-ng includes an on-the-spot picture feature, it has drawbacks.

A named topology diagram which was a screenshot of the eve-ng topology, which provides a visual help to speed up troubleshooting and recovery processes, improves these backups. It guarantees a successful resumption of service and is particularly beneficial for new team members or outside support.

This backup plan guarantees that Horizon Technologies is ready to handle interruptions and preserve the dependability and uninterrupted operation of its Internet service provider network.

Future work

Future plans for horizon aim for the deployment of a stand-alone network monitoring program to further improve the functionality and resilience of Horizon Technologies' ISP infrastructure. By identifying possible problems like bottlenecks, checking latency for each device in the network and device malfunctions, or security breaches before they affect operations, this tool would offer quick real-time insights into the horizon network performance. Because of their capacity to track network traffic patterns, bandwidth usage, and device health, applications such as SolarWinds Network Performance Monitor may be worth considering.

Using sophisticated layer 2 protocols like Rapid Spanning Tree Protocol (RSTP) will enhance network performance and avoid problems like broadcast storms in addition to monitoring for clients.

Enhancing security is another important priority for horizon, with plans to deploy next-generation firewalls for horizon and its clients and automated threat detection systems. These enhancements



would fortify defences against cyberattacks and enhance access management which in Bahrain mainly focuses on ISP networks and bank networks.

Traffic flow will be further optimised by increasing the usage of MPLS and QoS setups, guaranteeing that important data is given priority even during times of high traffic. Software-defined networking (SDN) will also be used to solve future scalability by making it easier to manage large-scale topologies and update configurations.

By strengthening the Horizon network's scalability, security, and performance, these planned future upgrades will guarantee that Horizon Technologies is prepared to handle expanding needs and changing obstacles throughout the company life cycle.

My experience

It has been a rewarding experience to design and construct Horizon Technologies' ISP network from the ground up, full with obstacles and worthwhile educational opportunities. A combination of technical expertise, problem-solving skills, and flexibility was needed throughout the process, from designing the network architecture to putting it into practice and debugging its different parts. I learnt the value of perseverance and comprehensive testing at every stage of the project from setbacks like corrupted EVE-NG photos and VRF connectivity problems, even if they also tried my patience.

From relearning the fundamental layer 2 protocols to grasping more complex ideas like MPLS, VRF, and IPsec, the experience improved my comprehension of networking technologies to implement in the Horizon network. Setting up devices, making sure there was redundancy, and resolving latency in a big topology improved my practical abilities and broadened my understanding of the challenges of implementing networks in the real world. Additionally, I discovered the need of thorough documentation and backups, which were crucial in regaining progress following setbacks.

In addition to increasing my technical proficiency and learning methods, this project taught me how to approach difficulties methodically and adjust to unforeseen obstacles like the ones listed in previous paragraphs. It was a fulfilling experience that increased my confidence in creating networks that are scalable, secure, and effective, setting me up for success in the networking and IT industries going forward.

Bahrain Perspective

The ISP network project from Horizon Technologies supports Bahrain's "Vision 2030" goals of technical advancement and economic diversification, and it is in line with the country's desire to become a regional technology hub. Reliable, scalable, and secure internet infrastructure is essential as Bahrain's digital economy expands to satisfy the growing needs of institutions, corporations, and individual users throughout the kingdom. By building a strong ISP infrastructure that facilitates seamless connectivity, improves data security, and guarantees effective resource management in a small kingdom like Bahrain, makes sure the project directly advances this goal.



In an economy like Bahrain's, where industries like finance, e-commerce, and IT services are expanding quickly, a well-designed ISP network is essential.

A well-designed ISP network is crucial in a rapidly rising economy like Bahrain's, where sectors like finance, e-commerce, and IT services are growing quickly.

Horizon technologies's network provides a backbone that facilitates advanced technologies, encourages digital transformation, and aids in the efficient operation of businesses. Additionally, making sure enhanced security features such as IPsec encryption and VRF for segmentation ensure data integrity and privacy, which is crucial for sectors like banking and healthcare and general corporate companies.

In the end, this project helps Bahrain achieve its goals of using cutting-edge technologies, promoting economic expansion, and solidifying its status as a pioneer in the region's digital revolution, all of which will help the Kingdom have a brighter, more connected future.

LESPI

Legal

Legally speaking, Horizon Technologies acts in accordance with Bahrain's telecom regulations and rules, which are set forth by the Telecommunications Regulatory Authority (TRA). According to Bahrain's Personal Data Protection Law (PDPL), this entails rigorous adherence to privacy and data protection regulations, guaranteeing the security and privacy of companies and user data.

Additionally, Horizon Technologies needs to fulfil the license standards established by the TRA for middle east, which regulate internet service providers in Bahrain. By explicitly stating its terms of service and privacy policies and for the user to accept, the business must uphold transparency with its users. Users can learn about their rights, the extent of data collection, and the use and protection of their information from these papers.

Ethical

Transparency, data security, and user privacy are given top priority by Horizon Technologies. The business makes sure that client information is safe, prevents misuse or illegal access, and is transparent about how it is collected and used.

Horizon Technologies maintains ethical accountability by tackling cybersecurity threats to protect clients in the country whether customers or companies and establish trust while promoting a safe



and welcoming digital environment for these customers. These actions promote a responsible digital ecosystem and are consistent with Bahrain's principles.

Professional

When providing ISP services, Horizon Technologies upholds the highest professional standards. The organisation uses cutting-edge protocols and industry best practices to guarantee dependable connectivity, strong security, and constant quality.

Clear communication with clients, prompt problem solving, and upholding operational transparency are all signs of professionalism. By emphasising innovation and quality, Horizon Technologies helps Bahrain move closer to having a cutting-edge, internationally competitive digital infrastructure.

Conclusion

The successful development and deployment of Horizon Technologies' ISP network in Bahrain represents a critical milestone in the construction of a reliable, secure, and expandable infrastructure that can accommodate the demands of on demand connections throughout the kingdom. Important networking issues were covered in this project, such as guaranteeing smooth internal and external communication between the ISP and companies themselves, putting advanced security measures in place, and maximizing system performance using cutting-edge technologies like MPLS, VRF, and IPsec.

Numerous challenges were overcome during the project, ranging from relearning basic networking protocols to debugging intricate settings in most protocols used. In addition to improving technical proficiency and solving problems quickly and efficiently, these experiences showed how crucial flexibility, careful planning, and the incorporation of dependable backup plans are to guarantee project durability especially in a big project like an ISP network.

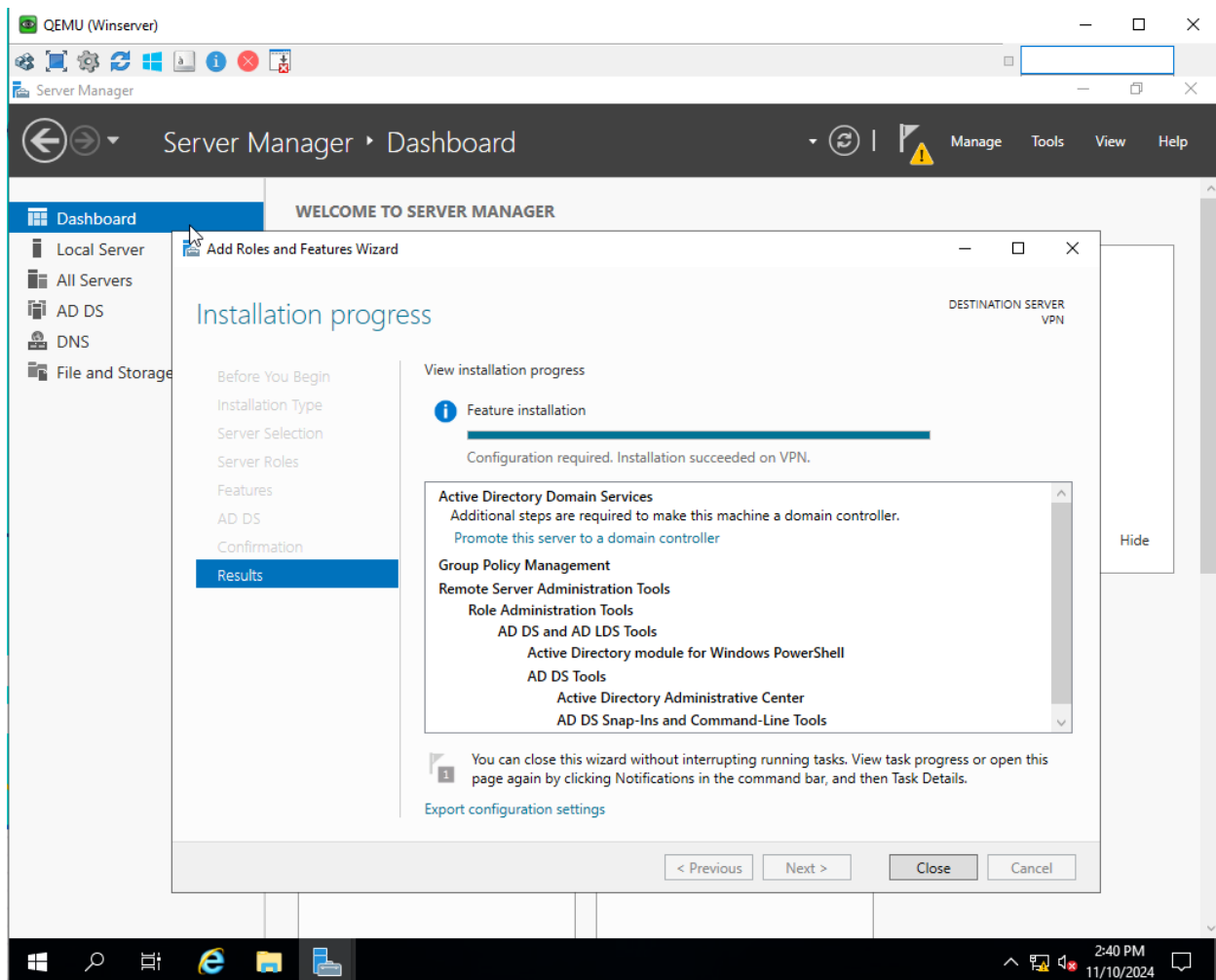
By fostering the development of safe and effective infrastructure and advancing the digital economy, this project supports Bahrain's "Vision 2030" from a national standpoint. This network puts Horizon Technologies in a position to satisfy the changing needs of people and organisations by addressing connection, scalability, and security, which will eventually promote innovation and advancement in the Kingdom.

This thesis demonstrates the value of a comprehensive strategy for ISP infrastructure that incorporates technical accuracy, adherence to the law, moral responsibility, and professional integrity. It draws attention to the possibilities for upcoming developments and future work in the company, such as the incorporation of monitoring systems for Horizon and improved scaling solutions in case customers get higher, guaranteeing Horizon Technologies' continued leadership in technical advancement in the kingdom of Bahrain and the middle east.

References

- Administrator. (n.d.). Configuring IPsec VPN on Cisco routers: A step-by-step guide.
<https://www.networkstraining.com/configuring-ipsec-vpn-on-cisco-routers/>
- Cisco Learning Library - Virtual routing and forwarding configuration. (n.d.).
https://www.cisco.com/resources/config_vrf.html
- ISACA. (n.d.). Professional ethics and standards for IT professionals.
<https://www.isaca.org/professional-ethics>
- Networking How-To Guides. (2023, November 18). Setting up a DHCP server on Cisco devices.
<https://www.networkinghowto.com/cisco-dhcp-server-setup/>
- IEEE Standards Association. (n.d.). IEEE Code of Conduct and Ethics.
<https://standards.ieee.org/code-of-conduct>
- Cisco Networking Guide - BGP Configuration for Beginners. (2020, October 22). Cisco.
<https://www.cisco.com/solutions/bgp-configuration-guide.html>
- Bahrain Telecommunications Regulatory Authority. (n.d.). Regulations and standards for ISPs in Bahrain. <https://www.tra.org.bh/isp-regulations>
- Packet Tracer Lab Exercises. (n.d.). How to configure MPLS in Cisco Packet Tracer.
<https://www.packettracerlab.com/mpls-configuration-guide/>
- Cisco Documentation - QoS and MPLS implementation. (2021, March 14). Cisco.
<https://www.cisco.com/docs/qos-and-mpls-guide.html>
- IT Governance Bahrain. (n.d.). Data protection and privacy laws in Bahrain.
<https://www.itgovernancebh.com/data-protection-laws/>
- SolarWinds. (2022, September 30). Network monitoring tools for ISPs: A guide.
<https://www.solarwinds.com/network-monitoring-guide>
- Computer Networking Insights. (2023, January 15). Troubleshooting VRF connectivity issues.
<https://www.networkinsights.com/troubleshooting-vrf/>
- TechRepublic. (2021, August 10). Configuring redundancy with EtherChannel in Cisco devices.
<https://www.techrepublic.com/etherchannel-configuration-guide>

Appendices



I installed using dashboard tools

Active directory

Dns services



QEMU (Winserver)

Server Manager

Server Manager ▸ Local Server

Dashboard

Local Server

All Servers

AD DS

DNS

File and Storage

Active Directory Domain Services Configuration Wizard

PROPERTIES For VPN

Paths

Deployment Configuration

Domain Controller Options

DNS Options

Additional Options

Paths

Review Options

Prerequisites Check

Installation

Results

Specify the location of the AD DS database, log files, and SYSVOL

Database folder: C:\Windows\NTDS

Log files folder: C:\Windows\NTDS

SYSVOL folder: C:\Windows\SYSVOL

TARGET SERVER VPN

Never Download Never

Real-Time Settings On (UTC-08:00) Not active

QEMU Virt 4 GB 39.46 GB

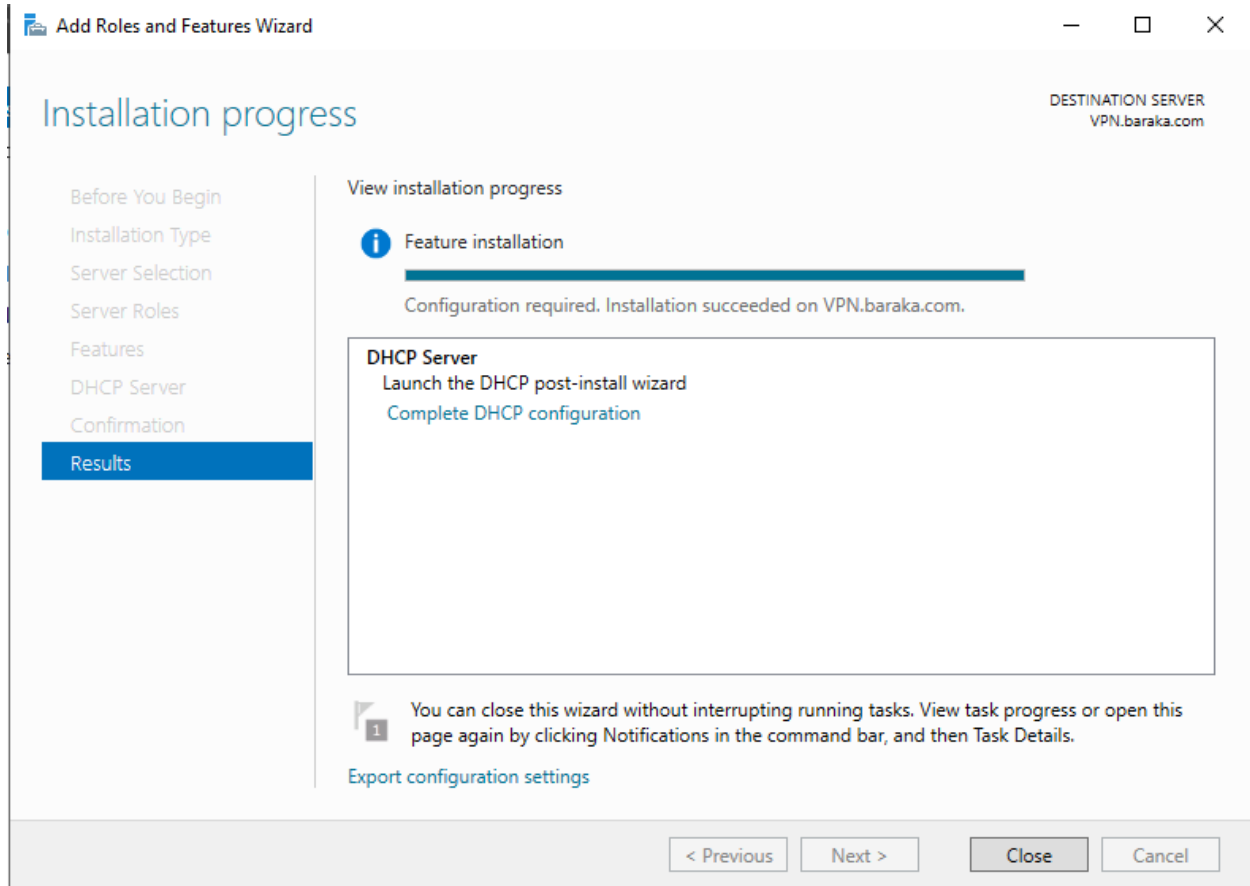
TASKS

More about Active Directory paths

< Previous Next > Install Cancel

VPN 1014 Error Microsoft-Windows-Security-SPP Application 11/10/2024 2:36:06 PM

2:45 PM 11/10/2024





HORIZON

New Scope Wizard

IP Address Range

You define the scope address range by identifying a set of consecutive IP addresses.



Configuration settings for DHCP Server

Enter the range of addresses that the scope distributes.

Start IP address:

End IP address:

Configuration settings that propagate to DHCP Client

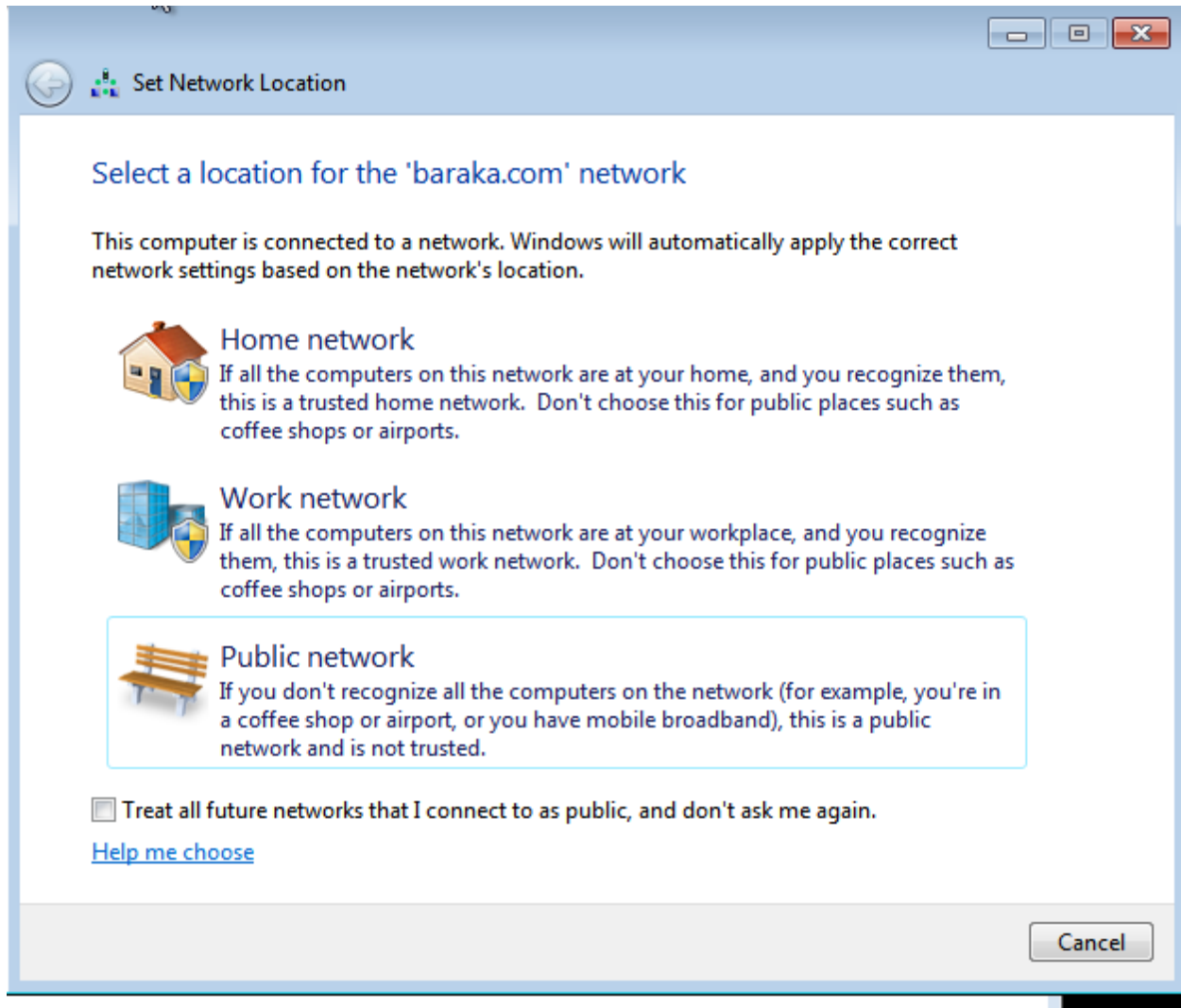
Length:

Subnet mask:

< Back

Next >

Cancel



R7 show run

R7(config)#do show run

Building configuration...

Current configuration : 1804 bytes

!

version 15.5

Page 67 of 94



HORIZON

service timestamps debug datetime msec

service timestamps log datetime msec

no service password-encryption

!

hostname R7

!

boot-start-marker

boot-end-marker

!

!

!

no aaa new-model

!

!

!

bsd-client server url <https://cloudsso.cisco.com/as/token.oauth2>

clock timezone EET 2 0

mmi polling-interval 60

no mmi auto-configure

no mmi pvc

mmi snmp-timeout 180

!

!

!

!

!

!

!



HORIZON

!

!

!

!

!

!

!

!

!

!

ip cef

no ipv6 cef

!

multilink bundle-name authenticated

!

!

!

!

!

!

!

cts logging verbose

!

!

!

redundancy



HORIZON

!

!

!

!

!

!

!

!

!

!

!

!

!

!

!

interface Ethernet0/0

no ip address

shutdown

!

interface Ethernet0/1

no ip address

shutdown

!

interface Ethernet0/2

no ip address

shutdown

!

interface Ethernet0/3



HORIZON

no ip address

shutdown

!

interface Serial1/0

ip address 30.0.0.1 255.255.255.252

serial restart-delay 0

!

interface Serial1/1

no ip address

shutdown

serial restart-delay 0

!

interface Serial1/2

ip address 30.0.0.5 255.255.255.252

serial restart-delay 0

!

interface Serial1/3

ip address 30.0.0.9 255.255.255.252

serial restart-delay 0

!

interface Serial2/0

no ip address

shutdown

serial restart-delay 0

!

interface Serial2/1

no ip address

shutdown



```
serial restart-delay 0
```

```
!
```

```
interface Serial2/2
```

```
no ip address
```

```
shutdown
```

```
serial restart-delay 0
```

```
!
```

```
interface Serial2/3
```

```
no ip address
```

```
shutdown
```

```
serial restart-delay 0
```

```
!
```

```
router bgp 200
```

```
bgp log-neighbor-changes
```

```
network 30.0.0.0 mask 255.255.255.252
```

```
network 30.0.0.4 mask 255.255.255.252
```

```
network 30.0.0.8 mask 255.255.255.252
```

```
neighbor 30.0.0.2 remote-as 100
```

```
neighbor 30.0.0.6 remote-as 300
```

```
neighbor 30.0.0.10 remote-as 400
```

```
!
```

```
ip forward-protocol nd
```

```
!
```

```
!
```

```
no ip http server
```

```
no ip http secure-server
```

```
!
```

```
!
```



```
!  
!  
control-plane  
!  
!  
!  
!  
!  
!  
!  
!  
line con 0  
  logging synchronous  
line aux 0  
line vty 0 4  
  login  
  transport input none  
!  
!  
End
```

R18 show run

```
Router(config)#do show run  
Building configuration...
```

Current configuration : 2039 bytes

```
!  
version 15.5  
Page 73 of 94
```



HORIZON

service timestamps debug datetime msec

service timestamps log datetime msec

no service password-encryption

!

hostname Router

!

boot-start-marker

boot-end-marker

!

!

!

no aaa new-model

!

!

!

bsd-client server url <https://cloudsso.cisco.com/as/token.oauth2>

clock timezone EET 2 0

mmi polling-interval 60

no mmi auto-configure

no mmi pvc

mmi snmp-timeout 180

!

!

!

!

!

!

!



HORIZON

!

!

!

!

!

!

!

ip vrf BLUE

rd 1:1

route-target export 1:1

route-target import 1:1

!

!

!

!

ip cef

no ipv6 cef

!

multilink bundle-name authenticated

!

!

!

!

!

!

!



HORIZON

cts logging verbose

!

!

!

redundancy

!

!

!

!

!

!

!

!

!

!

!

!

!

!

!

interface Ethernet0/0

ip vrf forwarding BLUE

ip address 50.0.0.30 255.255.255.252

!

interface Ethernet0/1

ip vrf forwarding BLUE

ip address 50.0.0.26 255.255.255.252

!



```
interface Ethernet0/2
ip vrf forwarding BLUE
ip address 70.0.0.1 255.255.255.252
ip access-group 101 in
!
interface Ethernet0/3
ip vrf forwarding BLUE
ip address 80.0.0.1 255.255.255.252
!
interface Ethernet1/0
ip vrf forwarding BLUE
ip address 90.0.0.1 255.255.255.252
!
interface Ethernet1/1
no ip address
shutdown
!
interface Ethernet1/2
no ip address
shutdown
!
interface Ethernet1/3
no ip address
shutdown
!
router ospf 1 vrf BLUE
router-id 18.18.18.18
redistribute static subnets
```

Page 77 of 94



HORIZON

network 50.0.0.24 0.0.0.3 area 0

network 50.0.0.28 0.0.0.3 area 0

network 70.0.0.0 0.0.0.3 area 0

network 80.0.0.0 0.0.0.3 area 0

network 90.0.0.0 0.0.0.3 area 0

!

ip forward-protocol nd

!

!

no ip http server

no ip http secure-server

!

!

!

access-list 101 deny ip host 192.168.1.5 192.168.2.0 0.0.0.255

access-list 101 permit ip any any

access-list 102 deny tcp 192.168.1.0 0.0.0.255 any eq www

access-list 102 deny tcp 192.168.1.0 0.0.0.255 any eq 443

access-list 102 deny tcp 192.168.1.0 0.0.0.255 any eq telnet

access-list 102 deny tcp 192.168.1.0 0.0.0.255 any eq smtp

access-list 102 permit ip any any

!

control-plane

!

!

!

!

!



```
!  
!  
!  
line con 0  
  logging synchronous  
line aux 0  
line vty 0 4  
  login  
  transport input none  
!  
!  
End
```

R20 show run

```
Router(config)#do show run  
Building configuration...
```

Current configuration : 1884 bytes

```
!  
version 15.5  
service timestamps debug datetime msec  
service timestamps log datetime msec  
no service password-encryption  
!  
hostname Router  
!  
boot-start-marker  
boot-end-marker  
Page 79 of 94
```



HORIZON

!

!

!

no aaa new-model

!

!

!

bsd-client server url <https://cloudsso.cisco.com/as/token.oauth2>

clock timezone EET 2 0

mmi polling-interval 60

no mmi auto-configure

no mmi pvc

mmi snmp-timeout 180

!

!

!

!

!

!

!

!

!

!

!

!

!



HORIZON

!

!

!

!

ip cef

no ipv6 cef

!

multilink bundle-name authenticated

!

!

!

!

!

!

!

cts logging verbose

!

!

!

redundancy

!

!

!

!

!

!

!

crypto isakmp policy 10



```
encr aes
hash sha256
authentication pre-share
group 14
crypto isakmp key vpn-shared-key address 50.0.0.54
!
!
crypto ipsec transform-set trans-set esp-aes esp-sha-hmac
mode tunnel
!
!
!
crypto map vpn-map 10 ipsec-isakmp
set peer 50.0.0.54
set transform-set trans-set
match address 100
!
!
!
!
!
interface Ethernet0/0
no ip address
shutdown
!
interface Ethernet0/1
no ip address
shutdown
```

Page 82 of 94



```
!  
interface Ethernet0/2  
  ip address 50.0.0.42 255.255.255.252  
  crypto map vpn-map  
!  
interface Ethernet0/3  
  no ip address  
!  
interface Ethernet0/3.10  
  encapsulation dot1Q 10  
  ip address 192.168.10.1 255.255.255.0  
!  
interface Ethernet1/0  
  no ip address  
!  
interface Ethernet1/0.20  
  encapsulation dot1Q 20  
  ip address 192.168.20.1 255.255.255.0  
!  
interface Ethernet1/1  
  no ip address  
  shutdown  
!  
interface Ethernet1/2  
  no ip address  
  shutdown  
!  
interface Ethernet1/3
```

Page 83 of 94



no ip address

shutdown

!

router ospf 50

network 50.0.0.40 0.0.0.3 area 0

network 192.168.10.0 0.0.0.255 area 0

network 192.168.20.0 0.0.0.255 area 0

!

ip forward-protocol nd

!

!

no ip http server

no ip http secure-server

!

ip access-list extended vpn-traffic

!

!

!

access-list 100 permit ip 50.0.0.40 0.0.0.3 60.0.0.0 0.0.0.255

!

control-plane

!

!

!

!

!

!

!



```
!  
line con 0  
  logging synchronous  
line aux 0  
line vty 0 4  
  login  
  transport input none  
!  
!  
End
```

R21 (switch) show run

```
Switch(config)#do show run  
Building configuration...  
  
Current configuration : 1095 bytes  
!  
version 15.1  
service timestamps debug datetime msec  
service timestamps log datetime msec  
no service password-encryption  
service compress-config  
!  
hostname Switch  
!  
boot-start-marker  
boot-end-marker  
!  
Page 85 of 94
```



HORIZON

!

!

no aaa new-model

clock timezone EET 2 0

!

ip cef

!

!

no ipv6 cef

ipv6 multicast rpf use-bgp

!

!

!

!

!

!

!

spanning-tree mode pvst

spanning-tree extend system-id

!

!

!

!

vlan internal allocation policy ascending

!

!

!

!



HORIZON

!

!

!

!

!

!

interface Port-channel1

switchport

switchport trunk encapsulation dot1q

switchport mode trunk

!

interface Ethernet0/0

switchport trunk encapsulation dot1q

switchport mode trunk

duplex auto

channel-group 1 mode active

!

interface Ethernet0/1

switchport trunk encapsulation dot1q

switchport mode trunk

duplex auto

!

interface Ethernet0/2

switchport access vlan 10

switchport mode access

duplex auto

!

interface Ethernet0/3



HORIZON

```
switchport trunk encapsulation dot1q
```

```
switchport mode trunk
```

```
duplex auto
```

```
channel-group 1 mode active
```

```
!
```

```
!
```

```
no ip http server
```

```
!
```

```
!
```

```
!
```

```
!
```

```
!
```

```
control-plane
```

```
!
```

```
!
```

```
line con 0
```

```
logging synchronous
```

```
line aux 0
```

```
line vty 0 4
```

```
login
```

```
!
```

```
End
```

R31 show run

```
Router(config)#do show run
```

```
Building configuration...
```

Current configuration : 1648 bytes

Page 88 of 94



```
!  
version 15.5  
service timestamps debug datetime msec  
service timestamps log datetime msec  
no service password-encryption  
!  
hostname Router  
!  
boot-start-marker  
boot-end-marker  
!  
!  
!  
no aaa new-model  
!  
!  
!  
bsd-client server url https://cloudsso.cisco.com/as/token.oauth2  
clock timezone EET 2 0  
mmi polling-interval 60  
no mmi auto-configure  
no mmi pvc  
mmi snmp-timeout 180  
!  
!  
!  
!  
!
```



HORIZON

!

!

!

!

!

!

!

!

!

!

!

!

ip cef

no ipv6 cef

!

multilink bundle-name authenticated

!

!

!

!

!

!

!

cts logging verbose

!

!



HORIZON

!

redundancy

!

!

!

!

!

!

!

crypto isakmp policy 10

encr aes

hash sha256

authentication pre-share

group 14

crypto isakmp key vpn-shared-key address 50.0.0.42

!

!

crypto ipsec transform-set trans-set esp-aes esp-sha-hmac

mode tunnel

!

!

!

crypto map vpn-map 10 ipsec-isakmp

set peer 50.0.0.42

set transform-set trans-set

match address 100

!

!



```
!  
!  
!  
interface Ethernet0/0  
  ip address 50.0.0.54 255.255.255.252  
  crypto map vpn-map  
!  
interface Ethernet0/1  
  ip address 50.0.0.58 255.255.255.252  
!  
interface Ethernet0/2  
  ip address 60.0.0.1 255.255.255.0  
!  
interface Ethernet0/3  
  no ip address  
  shutdown  
!  
router ospf 50  
  network 50.0.0.52 0.0.0.3 area 0  
  network 50.0.0.56 0.0.0.3 area 0  
  network 60.0.0.0 0.0.0.255 area 0  
!  
ip forward-protocol nd  
!  
!  
no ip http server  
no ip http secure-server  
!  
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```



HORIZON

```
ip access-list extended vpn-traffic
```

```
!
```

```
!
```

```
!
```

```
access-list 100 permit ip 50.0.0.52 0.0.0.3 192.168.10.0 0.0.0.255
```

```
access-list 130 permit ip 60.0.0.0 0.0.0.255 192.168.10.0 0.0.0.255
```

```
access-list 130 permit ip any any
```

```
!
```

```
control-plane
```

```
!
```

```
!
```

```
!
```

```
!
```

```
!
```

```
!
```

```
!
```

```
!
```

```
line con 0
```

```
logging synchronous
```

```
line aux 0
```

```
line vty 0 4
```

```
login
```

```
transport input none
```

```
!
```

```
!
```

```
end
```

